

Projective Geometry

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Introduction

Projective geometry is a branch of geometry. It is quite different from Euclidean geometry in the way it sets up the space. In Euclidean geometry, lengths and angles are among the most important things that we consider. However, these no longer become the vital parts in projective geometry. Indeed, there is no quantitative measure in projective geometry. What we are concerned about are simply points and lines.

In projective geometry, we mainly investigate some projections of points and lines. Intuitively, such transformations are carried out by projecting some points (or lines) from one surface to another surface. By using different centres and different surfaces for projections, we can obtain different relationships between the objects and their images. The whole topic of projective geometry is to study those special properties.

Our purpose of this note is to apply projective geometry to solve some geometric problems in Olympiad. Therefore, our main focus will be the projective plane instead of the general space (though working on a projective space of higher dimension could also solve some planar problems). Also, we are not going to introduce the projective space formally, i.e. we shall not state its fundamental axioms one by one. As Olympiad problems mainly involve Euclidean geometry, we shall try to connect the two things together and see how projective geometry can be applied to solve a problem. So one may take this as an additional tool for solving geometry problems.

In section 1, we first introduce the important concepts in projective geometry and the tools we shall use. We also include a few famous theorems in both projective geometry and Euclidean geometry and some of their applications. Then in section 2, we shall go through a special group of four points called a harmonic division together with its many properties. This has a lot of relations with Euclidean geometry, and hence is very useful for our purpose.

In addition, we shall introduce the pole and polar transformation in section 3. This is a transformation with respect to a conic, which has some usages in problems with a circle. The last section 4 is a short section about involution.

1 Projective Plane

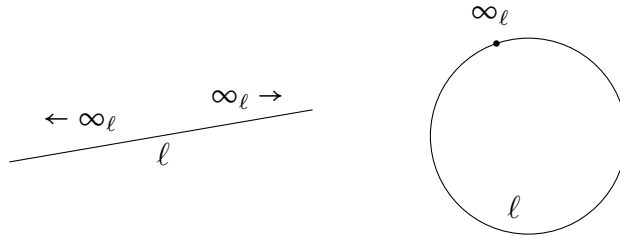
We are all supposed to be familiar with the Euclidean plane. However, the plane used in projective geometry is quite different. In this note, we shall try to define the projective plane by extending the usual Euclidean plane. Although this is not its exact definition, it is easier for those who have no knowledge in projective geometry to understand.

1.1 Concept of Infinity

The major difference between the projective plane and the Euclidean plane is the concept of infinity. A projective plane can be viewed as the usual Euclidean plane together with some *points at infinity*(無窮遠點) (or *ideal points*) and a line at infinity.

Firstly, we define a unique point at infinity for each direction. This means any line ℓ contains the point at infinity of its direction. In this note, we shall use ∞_ℓ to denote that point associated to the direction of ℓ . Note that the notation ∞ is vague as there are many different points at infinity instead of one. This is a bit different from some of the other extensions of the Euclidean plane. Indeed, two lines ℓ_1 and ℓ_2 contain the same point at infinity if and only if (iff) $\ell_1 // \ell_2$ in the sense of Euclidean geometry.

For each straight line ℓ , its point at infinity lies on either ‘end’ of the line. Although there seems to be two ‘ends’ for an ordinary line, we consider both ways to infinity to be the same. No matter which direction we travel, we will arrive at the same point at infinity. It may be easier to understand by viewing a line as a closed loop. This can be done by identifying the two ‘ends’ to the same point at infinity. Under this convention, we see that there is no difference between an ordinary point and an ideal point. This helps eliminate various cases.



Next, we define the *line at infinity*(無窮遠線) (or *ideal line*) to be the line containing all the ideal points. In particular, there is only one line at infinity.

Although we have the special names for ideal points and the ideal line, they behave the same as the ordinary ones on the projective plane. For example, we have the following facts.

- (a) For any two distinct points, there is a unique line joining them.
- (b) For any two distinct lines, there is a unique point lying on both of them.

Note that (a) is also valid in Euclidean geometry while (b) does not. Indeed, this fails to hold when the two lines are parallel. However, in projective geometry, both statements are valid in all cases and we do not have the concept of parallelism. Two ‘parallel’ lines indeed meet at a unique point, the point at infinity in their common direction.

In this note, we shall use $\ell_1 \cap \ell_2$ to denote the intersection point of the two distinct lines ℓ_1 and ℓ_2 , which could well be a point at infinity. We say that three points are *collinear*(共線) iff they lie on a single line (including the line at infinity), and three lines are *concurrent*(共點) iff they intersect at a single point (including a point at infinity). Note that the concept of concurrency includes parallelism.

Apart from the concept of infinity, there is another special feature of the projective plane, called the *principle of duality*(對偶原則). In projective geometry, points and lines are indistinguishable. That means the same fact usually holds when points and lines interchange their roles. For example, a point lying on a line is dual to a line passing through a point. Also, the two facts (a) and (b) given above are dual to each other. There will be more examples given later.

In projective geometry, we do not have any measurements like lengths or angles. However, for our purpose, we aim at connecting these things to the projective plane so that the facts learnt can be applied to solve problems in the Euclidean plane. In many cases, we shall use directed lengths instead of the usual lengths.

Directed lengths are only defined in the Euclidean plane, and so the concept of infinity is not considered. However, it is common to compute the ratio of directed lengths in projective geometry. In view of this, we can still compute such ratios if some points at infinity are involved. For example, if A and B are two ordinary points on ℓ , then $\frac{A\infty_\ell}{B\infty_\ell} = 1$, since the ratio $\frac{AP}{BP}$ tends to 1 when P moves towards ∞_ℓ on ℓ .

1.2 Pencils of Points and Lines

In this subsection, we shall introduce two important transformations on the projective plane, namely, perspective and projective transformations.

A *pencil of points*(點列) on a line ℓ is a collection of some points (usually finitely many) on the line. We use $\ell(P_1, P_2, P_3, \dots)$ to denote a pencil of points on ℓ , where

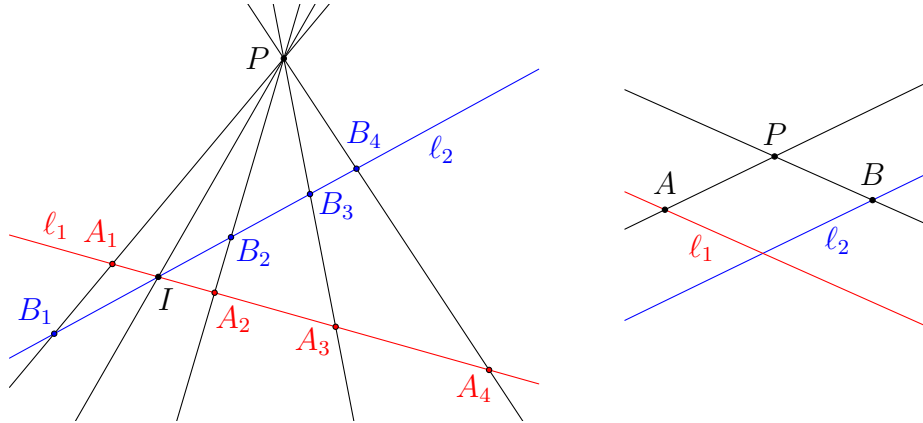
$P_1, P_2, P_3, \dots$ may lie on ℓ in any order. Sometimes we may abbreviate this as $(P_1, P_2, P_3, \dots)$ since it is clear which line the points lie on.

Dually, a *pencil of lines* (線束) on a point P is a collection of some lines passing through that point. We use $P(\ell_1, \ell_2, \ell_3, \dots)$ to denote a pencil of lines on P , provided that $\ell_1, \ell_2, \ell_3, \dots$ all pass through P . Again, we may drop the label P for simplicity.

Now, we can define the two transformations.

Definition 1.1. Let $\ell_1(A_1, A_2, A_3, \dots)$ be a pencil of points on a line ℓ_1 , let P be a point not on ℓ_1 , and let ℓ_2 be another line not passing through P . We define a *perspective transformation* (透視變換) from $\ell_1(A_1, A_2, A_3, \dots)$ to $\ell_2(B_1, B_2, B_3, \dots)$ with centre P to be a mapping such that the image of each A_j is the unique point $B_j = PA_j \cap \ell_2$.

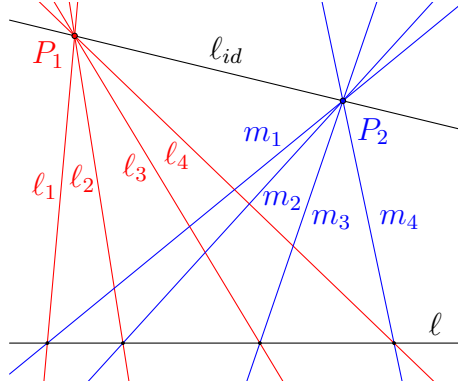
In this note, we use the notation $\ell_1(A_1, A_2, A_3, \dots) \xrightarrow{P} \ell_2(B_1, B_2, B_3, \dots)$ to denote the transformation. We say that the two pencils $\ell_1(A_1, A_2, A_3, \dots)$ and $\ell_2(B_1, B_2, B_3, \dots)$ are *perspective* (透視) to each other. Note that there is a unique point $I = \ell_1 \cap \ell_2$ which is invariant under this transformation.



If A is a point on ℓ_1 such that $PA \parallel \ell_2$, the image of A is the point at infinity of ℓ_2 . Similarly, the image of the point at infinity of ℓ_1 is the point B on ℓ_2 for which $PB \parallel \ell_1$.

Dually, we can define a perspectivity between two pencils of lines.

Definition 1.2. Let $P_1(\ell_1, \ell_2, \ell_3, \dots)$ be a pencil of lines on a point P_1 , let ℓ be a line not on P_1 , and let P_2 be another point not on ℓ . We define a *perspective transformation* (透視變換) from $P_1(\ell_1, \ell_2, \ell_3, \dots)$ to $P_2(m_1, m_2, m_3, \dots)$ through ℓ to be a mapping such that the image of each ℓ_j is the unique line m_j passing through $\ell_j \cap \ell$ and P_2 .



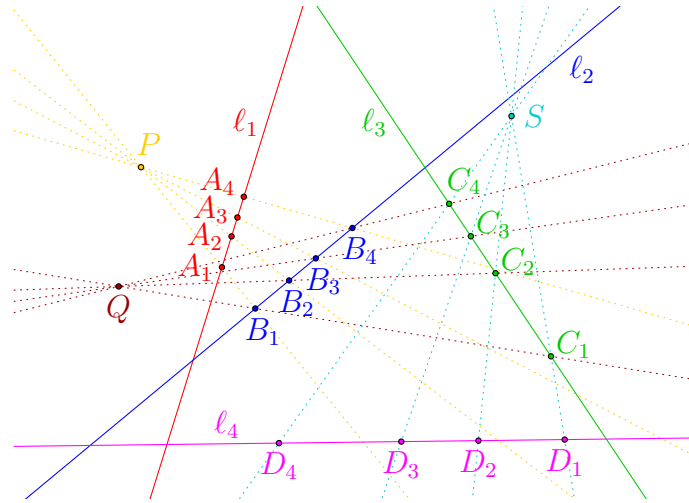
Similarly, we use the notation $P_1(\ell_1, \ell_2, \ell_3, \dots) \xrightarrow{\ell} P_2(m_1, m_2, m_3, \dots)$ to denote the transformation. We say that the two pencils $P_1(\ell_1, \ell_2, \ell_3, \dots)$ and $P_2(m_1, m_2, m_3, \dots)$ are *perspective* (透視) to each other. There is a unique line ℓ_{id} joining P_1 and P_2 which is invariant under this transformation.

For our purpose, we shall mainly use the perspectivities between pencils of points rather than pencils of lines. But once again, these are dual properties and can be interchanged easily.

Next, we define the following.

Definition 1.3. A *projective transformation* (射影變換) is a sequence of perspectivities.

Two pencils of points (lines) are said to be *projective* iff there exists a sequence of perspectivities mapping one of them to the other. Note that such a sequence is not unique, if exists. The following figure gives an example of a projectivity between $\ell_1(A_1, A_2, A_3, A_4)$ and $\ell_4(D_1, D_2, D_3, D_4)$ through the centres P, Q, S .



1.3 Cross Ratio

Cross ratio is the most useful tool that relates lengths in Euclidean geometry and objects in projective geometry.

Definition 1.4. Let A, B, C, D be four distinct points on a line ℓ in any order. The *cross ratio*(交比) (or *double ratio*) of A, B with respect to(w.r.t.) C, D is defined to be

$$(A, B; C, D) = \frac{\frac{AC}{AD}}{\frac{BC}{BD}} = \frac{AC \cdot BD}{AD \cdot BC}.$$

The lengths used are directed lengths.

As directed lengths are used, we should not mix up AC with CA , etc. Note that A, B, C, D could be ordinary points or points at infinity as we are only interested in the ratio of lengths. For example, if A is an ideal point, then $\frac{AC}{AD} = 1$, and hence

$$(A, B; C, D) = \frac{BD}{BC}.$$

By definition, we have

$$(A, B; C, D) = (B, A; D, C) = (C, D; A, B) = (D, C; B, A)$$

and $(B, A; C, D) = \frac{1}{(A, B; C, D)}$, etc. Thus, the cross ratio of four points mainly depends on how they are grouped into two pairs of points. This suggests the phrase we use in the definition as well as the notation. The importance of cross ratio is given in the following theorem.

Theorem 1.1. (*Cross ratio theorem*(交比定理)) The cross ratio of a pencil of four distinct points is invariant under a perspectivity.

Proof. By considering all possible configurations, we see that the sign of the cross ratio is preserved. Therefore, it suffices to show that the absolute value of the cross ratio remains unchanged. For simplicity, we only consider the case of finite points.

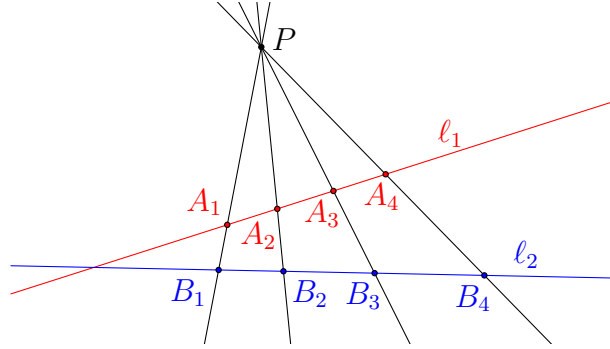
By the sine law, we have $A_1A_3 = \frac{PA_1 \sin \angle A_1PA_3}{\sin \angle PA_3A_1}$ and $A_1A_4 = \frac{PA_1 \sin \angle A_1PA_4}{\sin \angle PA_4A_1}$.

Therefore,

$$\frac{A_1A_3}{A_1A_4} = \frac{\sin \angle A_1PA_3 \sin \angle PA_4A_1}{\sin \angle PA_3A_1 \sin \angle A_1PA_4}.$$

Similarly, we obtain

$$\frac{A_2A_3}{A_2A_4} = \frac{\sin \angle A_2PA_3 \sin \angle PA_4A_2}{\sin \angle PA_3A_2 \sin \angle A_2PA_4}.$$



As $\sin \angle PA_3A_1 = \sin \angle PA_3A_2$ and $\sin \angle PA_4A_1 = \sin \angle PA_4A_2$ (which are true in any configuration), this gives

$$|(A_1, A_2; A_3, A_4)| = \frac{\sin \angle A_1PA_3 \sin \angle A_2PA_4}{\sin \angle A_1PA_4 \sin \angle A_2PA_3}.$$

The expression for $|(B_1, B_2; B_3, B_4)|$ is obtained by replacing A by B . Note that the two expressions are the same as the angles at P are equal by the definition of perspectivity. The proof is complete. $\square$

As a projectivity is a sequence of perspectivities, the cross ratio of four points remains unchanged under projective transformations.

By the cross ratio theorem, we know that for any projectivity between two pencils $\ell_1(A_1, A_2, A_3, \dots)$ and $\ell_2(B_1, B_2, B_3, \dots)$, once the images of three distinct points, say A_1, A_2, A_3 , are known, the images of all other points are uniquely determined. Indeed, given fixed collinear points A, B, C , there is a unique point D on the same

line which satisfies a fixed cross ratio $(A, B; C, D) = \frac{\overline{AC}}{\overline{BC}} = \frac{\overline{AD}}{\overline{BD}}$, as the ratio $\frac{\overline{AD}}{\overline{BD}}$ uniquely determines the position of D (since directed lengths are used).

This motivates the consideration of how a pencil of three points behaves under a projection. We have the following fundamental theorem.

Theorem 1.2. Let $\ell_1(A_1, A_2, A_3)$ and $\ell_2(B_1, B_2, B_3)$ be two pencils of points (possibly with $\ell_1 = \ell_2$). Then there exists a unique projective transformation mapping A_1, A_2, A_3 to B_1, B_2, B_3 respectively.

Here, the uniqueness means the images are fixed, while the sequence of perspectivities is not unique.

Proof. From the above discussion, the uniqueness follows from the cross ratio theorem. For the existence, we shall consider a few possible cases.

Case 1. $\ell_1 \neq \ell_2$ and $A_1 = B_1$ (or $A_2 = B_2$, $A_3 = B_3$)

Let $P = A_2B_2 \cap A_3B_3$ be the centre of perspectivity. Then

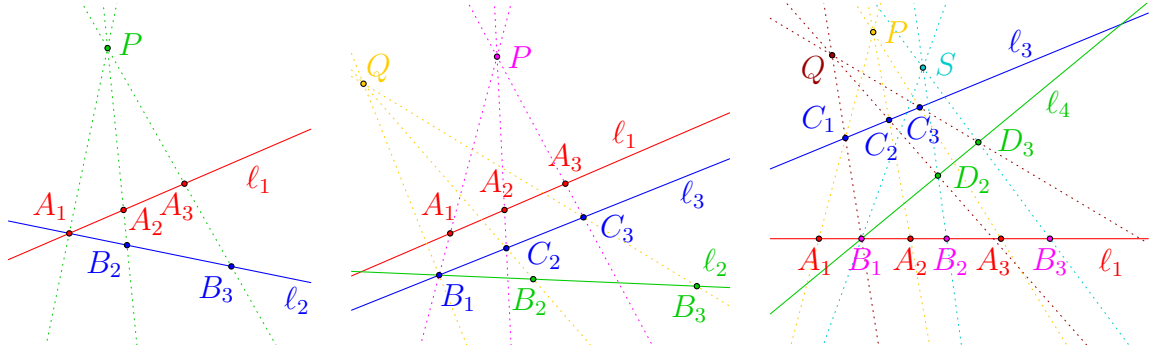
$$\ell_1(A_1, A_2, A_3) \xrightarrow{P} \ell_2(B_1, B_2, B_3).$$

Case 2. $\ell_1 \neq \ell_2$ and $A_j \neq B_j$ for $j = 1, 2, 3$

We first pick any point $P \neq A_1, B_1$ on A_1B_1 to be the centre and project A_1, A_2, A_3 to B_1, C_2, C_3 on some line $\ell_3 \neq \ell_2$. Then it turns into **case 1**.

Case 3. $\ell_1 = \ell_2$

In this case, we first take any point P not on ℓ_1 as centre and project A_1, A_2, A_3 to C_1, C_2, C_3 on some line ℓ_3 so that the points $B_1, B_2, B_3, C_1, C_2, C_3$ are distinct. It turns into **case 2**.



□

Using **theorem 1.2**, we can derive the following useful facts.

Proposition 1.1. The following results hold.

- (a) Any projective transformation which maps three collinear points back to themselves is the identity map.
- (b) A projective transformation from ℓ_1 to ℓ_2 (where $\ell_1 \neq \ell_2$) is a perspective transformation iff it maps $\ell_1 \cap \ell_2$ to itself.

Proof.

- (a) This is an immediate consequence of **theorem 1.2**.

- (b) Suppose $\ell_1(A_1, A_2, A_3, \dots)$ and $\ell_2(B_1, B_2, B_3, \dots)$ are projective to each other where $A_1 = \ell_1 \cap \ell_2$.

If it is a perspective transformation, then B_1 must be A_1 as any line through A_1 must intersect ℓ_2 at this point.

If $B_1 = A_1$, consider $P = A_2B_2 \cap A_3B_3$. The perspectivity from centre P clearly maps A_1, A_2, A_3 to B_1, B_2, B_3 respectively. By [theorem 1.2](#), the projection from $\ell_1(A_1, A_2, A_3, \dots)$ to $\ell_2(B_1, B_2, B_3, \dots)$ is uniquely determined by the images of A_1, A_2, A_3 . So this must be the perspective transformation.

□

Part (b) can be used to prove that three lines are concurrent. Indeed, if we can show that $\ell_1(A_1, A_2, A_3, A_4)$ and $\ell_2(B_1, B_2, B_3, B_4)$ are projective for some distinct lines ℓ_1, ℓ_2 where $A_1 = B_1$, then they must be perspective to each other, and hence A_2B_2, A_3B_3, A_4B_4 are concurrent at the centre.

Recall that all definitions and facts above have their dual forms when points and lines are interchanged. In particular, we can define the cross ratio of four lines through a point P to be the cross ratio of the four points cut by any line not on P . This is well-defined by the cross ratio theorem. Also, we have the following fact.

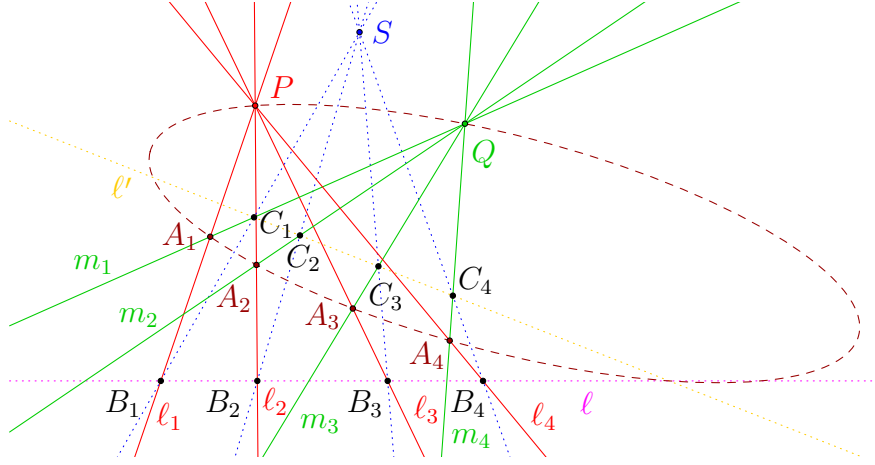
Proposition 1.2. A projective transformation from P_1 to P_2 (where $P_1 \neq P_2$) is a perspective transformation iff it maps P_1P_2 to itself.

1.4 Conic

In projective geometry, the definition of conics is quite different from that in Euclidean geometry.

We first define a point conic. Consider a non-perspective projection from a pencil of lines on a point P to a pencil of lines on another point Q . For each line ℓ_j on P and its image m_j on Q , we can form the intersection point $A_j = \ell_j \cap m_j$. The locus of all such points A_j is a certain shape on the plane called a [point conic](#) (圓錐曲線/二次曲線) (or simply a [conic](#)). It turns out that this definition is consistent to the usual definition of conics in the Euclidean plane, though we shall not prove this.

For example, the figure below shows a projectivity $P \xrightarrow{\ell} S \xrightarrow{\ell'} Q$ between P and Q . For each line PB_j , its image QC_j intersects it at the point A_j . The locus of all A_j 's is an ellipse as shown.



It could also happen that the point conic formed is a parabola or a hyperbola. The main difference between these kinds of conics is the number of points at infinity they contain. We have the following classification.

- An *ellipse*(橢圓) contains no point at infinity, and so it is a bounded curve.
- A *parabola*(拋物線) contains a single point at infinity, which lies on the axis of symmetry of the parabola.
- A *hyperbola*(雙曲線) contains two points at infinity, which lie on the two asymptotes of the hyperbola.

Note that a circle is a special case of an ellipse, and this is also the only kind of conics that we shall deal with in this note. We require the projectivity between P and Q to be non-perspective to get rid of the case of a pair of straight lines. However, most results for conics are still valid in this limiting case.

The points P and Q in the definition must lie on the conic. To see this, the image of the line PQ on P is another line on Q , which cannot be QP as the projectivity is not a perspectivity. Therefore, the two distinct lines intersect at Q , and Q lies on the conic by definition. Similarly, the preimage of the line QP on Q is a certain line on P , which intersects QP at P . So P lies on the conic as well.

By [theorem 1.2](#), a projective transformation is uniquely determined by three pairs of preimages and images. Therefore, if we are given distinct points P, Q, A_1, A_2, A_3 with no three collinear, then there exists a unique projectivity between P and Q which sends PA_1, PA_2, PA_3 to QA_1, QA_2, QA_3 respectively. It follows that a conic can be formed from these five points. In addition, one can show that the conic formed does not depend on the choice of the pair of points for the projectivity. In other words, any five distinct points with no three collinear uniquely determine a conic. For the degenerate cases where some three or more points are collinear, they might determine a pair of distinct straight lines or a pair of superposed lines, i.e. a single line.

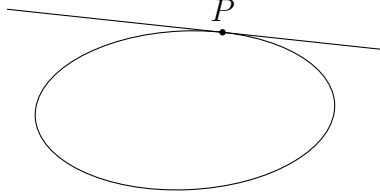
Next, we show a simple property of conics, which is a trivial fact if we use the definition in the Euclidean plane.

Proposition 1.3. Every straight line can intersect a conic in at most two points.

Proof. Suppose a straight line meets a conic at P and Q . We may suppose the conic is generated by a projective transformation between P and Q . It suffices to show that any point other than P and Q on PQ does not lie on the conic.

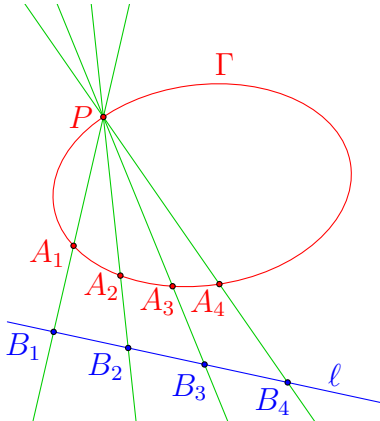
Suppose on the contrary that T lies on PQ and the conic. Then the image of PT on P is QT on Q . Both refer to the same line PQ , which cannot happen for a non-perspective projectivity by [proposition 1.1\(b\)](#). This leads to a contradiction. $\square$

As usual, any line which meets a conic in exactly one point is called a *tangent* (切線) to the conic. The point of intersection is called the contact point. For any point P on the conic, there exists a unique tangent at P to the conic. In particular, if the conic is determined by a projectivity between P and Q , then the image of PQ on P is the tangent at Q , while the preimage of QP on Q is the tangent at P .



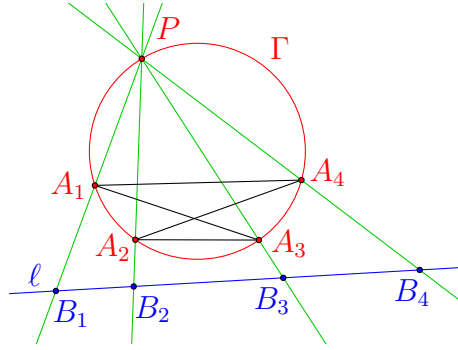
The dual of a point conic is a *line conic*. Instead of formed by the collection of points, a line conic is the shape enclosed by a collection of lines, which we shall not specify here. It is found that both definitions yield the same class of shapes.

Next, a *pencil of points* on a conic can be similarly defined. It refers to a collection of points on the conic Γ , say $\Gamma(P_1, P_2, P_3, \dots)$. We can form a *perspective transformation* between a pencil on a conic and a pencil on a line as well. Let $\Gamma(A_1, A_2, A_3, \dots)$ be a pencil on a conic and let $\ell(B_1, B_2, B_3, \dots)$ be a pencil on a line. They are said to be *perspective* iff $A_1B_1, A_2B_2, A_3B_3, \dots$ intersect at a point P on Γ . We denote the transformation by $\Gamma(A_1, A_2, A_3, \dots) \xrightarrow{P} \ell(B_1, B_2, B_3, \dots)$ as before.



Unlike a perspectivity between two lines, we require the centre of the perspectivity between a conic and a line to be lying on the conic. For the degenerate case when $A_1 = P$, the line PA_1 refers to the tangent at A_1 . Also, we do not have any perspective transformation between two conics. In a similar manner, we say that two pencils on conics or lines are *projective* if there exists a sequence of perspectivities between them, possibly including some perspectivities between lines.

Also, we can define the *cross ratio* of four points on a conic to be the cross ratio of the four points on a line which is perspective to them. From the definition of conics and the cross ratio theorem, we note that $(A, B; C, D)$ (where A, B, C, D lie on the conic) is irrespective of the centre of perspectivity or the line chosen. So it is well-defined. In addition, the cross ratio theorem can be extended to pencils on conics, i.e. the cross ratio of a pencil of four distinct points is invariant under projective transformations.



For the case of a circle, suppose $\Gamma(A_1, A_2, A_3, A_4) \xrightarrow{P} \ell(B_1, B_2, B_3, B_4)$ as shown. By definition, we have

$$(A_1, A_2; A_3, A_4) = (B_1, B_2; B_3, B_4) = \frac{\frac{B_1 B_3}{B_1 B_4}}{\frac{B_2 B_3}{B_2 B_4}}.$$

Using similar triangles, one can deduce

$$\frac{A_1 A_3}{B_1 B_3} = \frac{P A_1}{P B_3}, \quad \frac{A_1 A_4}{B_1 B_4} = \frac{P A_1}{P B_4}, \quad \frac{A_2 A_3}{B_2 B_3} = \frac{P A_2}{P B_3}, \quad \frac{A_2 A_4}{B_2 B_4} = \frac{P A_2}{P B_4}.$$

Therefore,

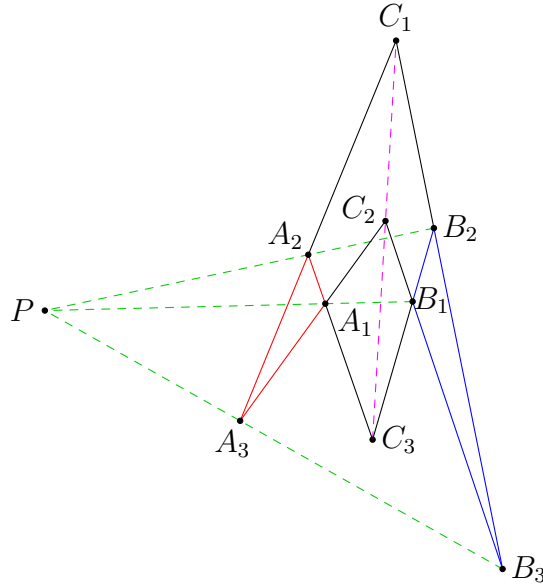
$$(A_1, A_2; A_3, A_4) = \frac{\frac{B_1 B_3}{B_1 B_4}}{\frac{B_2 B_3}{B_2 B_4}} = \frac{\frac{A_1 A_3}{A_1 A_4}}{\frac{A_2 A_3}{A_2 A_4}},$$

which is the same as the formula for the cross ratio of four points on a line. However, one should take care of the sign of the expression. One simple way of checking is that a negative sign is chosen iff A_1, A_2 are separated by A_3, A_4 on the circumference. Note that the formula is not true for general conics.

1.5 Projective Theorems

In this subsection, we shall go through a few famous theorems in projective geometry (which are also well-known in Euclidean geometry). Then we shall see how these theorems and the fundamental facts can be applied to solve problems in Euclidean geometry.

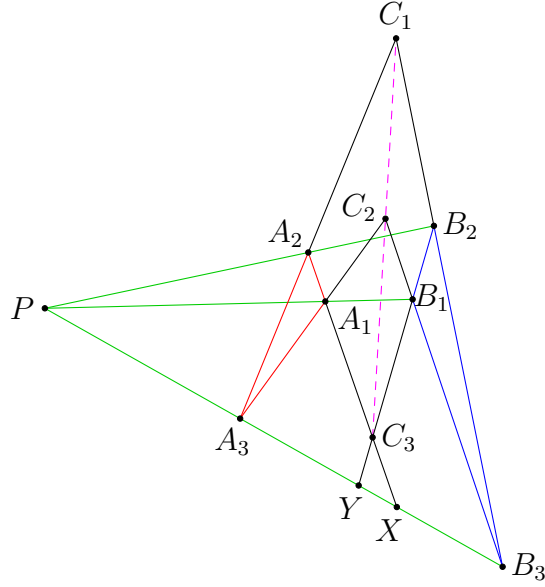
Theorem 1.3. (Desargues' theorem(笛沙格定理)) Given $\triangle A_1A_2A_3$ and $\triangle B_1B_2B_3$, let $C_1 = A_2A_3 \cap B_2B_3$, $C_2 = A_3A_1 \cap B_3B_1$ and $C_3 = A_1A_2 \cap B_1B_2$. Assume all points are well-defined and are distinct. Then A_1B_1, A_2B_2, A_3B_3 are concurrent iff C_1, C_2, C_3 are collinear.



We say that two triangles are *perspective*(透視) from a point if the three lines joining the corresponding vertices of the triangles are concurrent. Also, two triangles are *perspective* from a line if the three intersection points of corresponding sides of the triangles are collinear. Therefore, Desargues' theorem says that two triangles are perspective from a point iff they are perspective from a line.

Proof. If A_1B_1, A_2B_2, A_3B_3 are concurrent at P , we define $X = A_3B_3 \cap A_1A_2$ and $Y = A_3B_3 \cap B_1B_2$. Then we establish the following projective transformation.

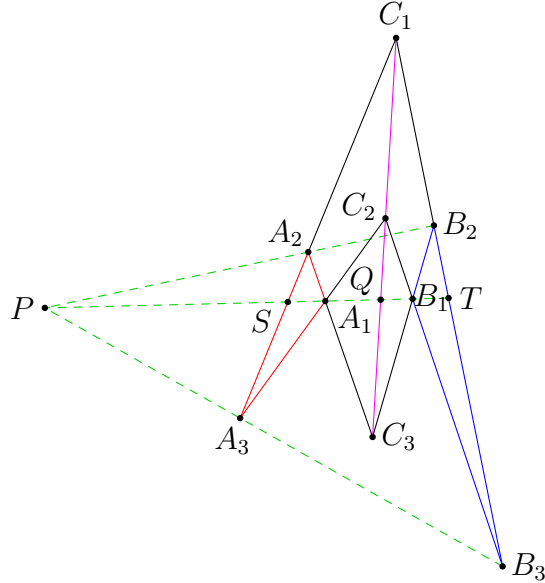
$$\begin{aligned} A_3(A_3C_1, A_3C_2, A_3B_3, A_3C_3) &\xrightarrow{A_2A_1} P(PA_2, PA_1, PX, PC_3) \\ &= P(PB_2, PB_1, PY, PC_3) \\ &\xrightarrow{B_2B_1} B_3(B_3C_1, B_3C_2, B_3A_3, B_3C_3) \end{aligned}$$



This shows $A_3(A_3C_1, A_3C_2, A_3B_3, A_3C_3)$ and $B_3(B_3C_1, B_3C_2, B_3A_3, B_3C_3)$ are projective. Note that they share a common line A_3B_3 . By [proposition 1.2](#), the two pencils are perspective. The corresponding intersection points $C_1 = A_3C_1 \cap B_3C_1$, $C_2 = A_3C_2 \cap B_3C_2$ and $C_3 = A_3C_3 \cap B_3C_3$ are thus collinear.

If C_1, C_2, C_3 are collinear, let A_1B_1 meet C_1C_2, A_2A_3, B_2B_3 at Q, S, T respectively. Then we can establish the following projection.

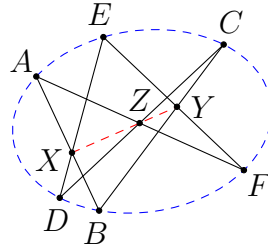
$$A_2A_3(A_3, S, A_2, C_1) \xrightarrow{A_1} C_1C_2(C_2, Q, C_3, C_1) \xrightarrow{B_1} B_2B_3(B_3, T, B_2, C_1)$$



Again, the two pencils $A_2A_3(A_3, S, A_2, C_1)$ and $B_2B_3(B_3, T, B_2, C_1)$ share a common point C_1 . By [proposition 1.1\(b\)](#), they are perspective. Therefore, A_3B_3, ST, A_2B_2 are concurrent, i.e. A_1B_1, A_2B_2, A_3B_3 are concurrent. $\square$

Desargues' theorem can be used to prove collinearity or concurrency. Note that it includes some limiting cases as well. For example, when A_1B_1, A_2B_2, A_3B_3 are parallel, the two triangles are perspective from a point at infinity. So they are perspective from a line. Similarly, when the corresponding sides of the two triangles are parallel, they are perspective from the line at infinity. In that case, the triangles are simply homothetic or congruent to each other, so they are perspective from a point.

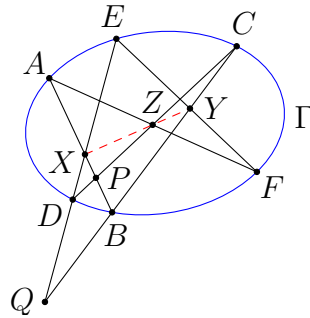
Theorem 1.4. ([Pascal's theorem](#)(帕斯卡定理)) Given points A, B, C, D, E, F , let $X = AB \cap DE$, $Y = BC \cap EF$ and $Z = CD \cap FA$. Assume all points are well-defined. Then X, Y, Z are collinear iff A, B, C, D, E, F lie on a conic.



Proof. Let $P = AB \cap CD$ and $Q = BC \cap DE$. If A, B, C, D, E, F lie on a conic Γ , then we have the following projection.

$$CD(D, P, Z, C) \xrightarrow{A} \Gamma(D, B, F, C) \xrightarrow{E} BC(Q, B, Y, C)$$

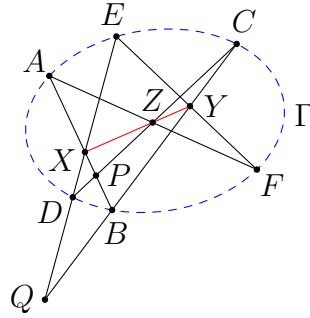
Since C is invariant, the two pencils (D, P, Z, C) and (Q, B, Y, C) are perspective by [proposition 1.1\(b\)](#). It follows that DQ, PB, ZY are concurrent at X , i.e. X, Y, Z are collinear.



If X, Y, Z are collinear, let Γ be the conic determined by A, B, C, D, E . Suppose AF and EF intersect Γ at F_1 and F_2 again respectively. Then we have

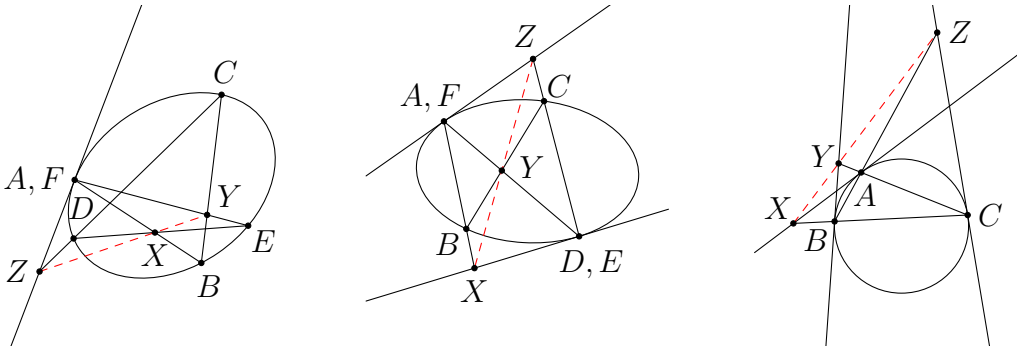
$$\Gamma(D, B, F_1, C) \xrightarrow{A} CD(D, P, Z, C) \xrightarrow{X} BC(Q, B, Y, C) \xrightarrow{E} \Gamma(D, B, F_2, C).$$

As D, B, C are invariants, [proposition 1.1\(a\)](#) ensures that this is the identity mapping. So $F_1 = F_2$, and in particular, both represent the point F . This means F lies on the conic determined by A, B, C, D, E .



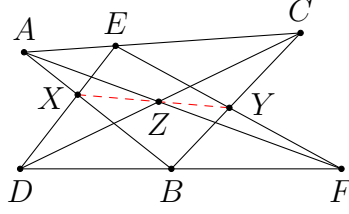
□

Since we shall only deal with circles, the ‘if’ part of Pascal’s theorem is more important. Also, this theorem consists of quite a lot of degenerate cases. It is allowed that some of the six points coincide, in which case the line joining the two coincide points refers to the tangent at the point to the conic. The following figures show some possible degenerate cases of Pascal’s theorem.



Theorem 1.5. ([Pappus’ theorem](#)(帕普斯定理)) Given points A, C, E on one line and points B, D, F on another line, let $X = AB \cap DE$, $Y = BC \cap EF$ and $Z = CD \cap FA$. Assume all points are distinct. Then X, Y, Z are collinear.

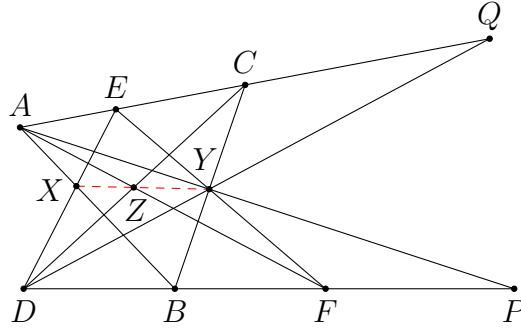
This is simply a special case of Pascal’s theorem, where the conic becomes a pair of straight lines. Nevertheless, we shall provide an alternative proof here.



Proof. Let $P = AY \cap BD$ and $Q = DY \cap AC$. Then we get the following projectivity.

$$A(AD, AX, AZ, AY) \xrightarrow{DB} Y(YD, YB, YF, YP)$$

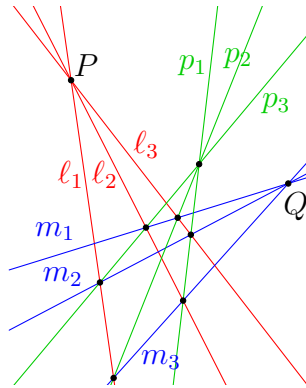
Similarly, we have $D(DA, DX, DZ, DY) \xrightarrow{AC} Y(YA, YE, YC, YQ)$.



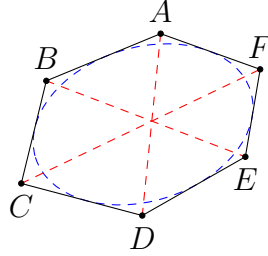
Note that the right-hand sides of the above two pencils contain the same set of four lines up to permutation. Indeed, by definition, the cross ratios $(YD, YB; YF, YP)$ and $(YP, YF; YB, YD)$ are equal. Therefore, the pencils (YD, YB, YF, YP) and (YP, YF, YB, YD) are projective to each other.

From this, we know that (AD, AX, AZ, AY) and (DA, DX, DZ, DY) are projective. Noting the invariance AD and applying [proposition 1.2](#) as before, we obtain the result that $AX \cap DX = X$, $AZ \cap DZ = Z$ and $AY \cap DY = Y$ are collinear. $\square$

There is a dual version of Pappus' theorem. Given lines ℓ_1, ℓ_2, ℓ_3 on one point and lines m_1, m_2, m_3 on another point, let p_1 be the line joining $\ell_2 \cap m_3$ and $\ell_3 \cap m_2$, let p_2 be the line joining $\ell_1 \cap m_3$ and $\ell_3 \cap m_1$, and let p_3 be the line joining $\ell_1 \cap m_2$ and $\ell_2 \cap m_1$. Then p_1, p_2, p_3 are concurrent.



Theorem 1.6. ([Brianchon's theorem](#)(布列安桑定理)) Given a hexagon $ABCDEF$, its main diagonals AD, BE, CF are concurrent iff it is circumscribed about a conic.



This is the dual of Pascal's theorem. We shall provide a proof in section 3. Brianchon's theorem has fewer applications than Pascal's theorem in Olympiad.

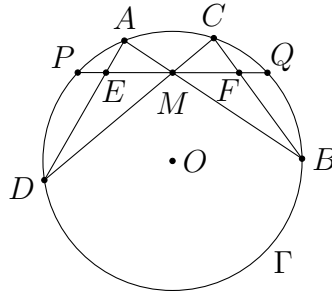
Next, we shall go through some problems that can be solved using the things introduced in this section. Firstly, we have some applications of cross ratios.

Problem 1.1. ([Butterfly theorem](#)(蝴蝶定理)) Given a circle Γ with centre O , let M be the midpoint of a chord PQ of Γ . Two other chords passing through M intersect Γ at A, B and C, D respectively. Let PQ meet AD and BC at E and F respectively. Prove that $ME = MF$.

Analysis. To derive some information from cross ratios, it means to establish a projectivity between two pencils of four points, so that we know their cross ratios are equal. From this, we can obtain a relationship between some lengths, which might yield the result at once, or provide useful information that can be incorporated into other geometric tools.

For this problem, since both ME and MF are segments on the chord PQ , we should probably deal with some pencils on this line. From $MP = MQ$ and $ME = MF$ (which have to be true), there are some pairs of quadruples having the same cross ratio (like those symmetric to each other). Also, starting from any pencil containing E, M, F does not seem to work as the projective step is hard to carry out without adding new points. So our choice in the following proof is reasonable.

Solution.



Consider the pencil $PQ(P, E, M, Q)$. To form a perspectivity, as there are only two lines containing E (with one being PQ), the centre of perspectivity can only be A or D . Both will lead to the same result. Thus, we easily get the following.

$$PQ(P, E, M, Q) \xrightarrow{A} \Gamma(P, D, B, Q) \xrightarrow{C} PQ(P, M, F, Q)$$

By the cross ratio theorem, this implies $(P, E; M, Q) = (P, M; F, Q)$. Now,

$$\begin{aligned} (P, E; M, Q) &= (P, M; F, Q) \\ \Rightarrow \frac{\frac{PM}{PQ}}{\frac{EM}{EQ}} &= \frac{\frac{PF}{PQ}}{\frac{MF}{MQ}} \\ \Rightarrow \frac{PM(EM + MQ)}{PQ \cdot EM} &= \frac{MQ(PM + MF)}{PQ \cdot MF} \\ \Rightarrow \frac{EM + MQ}{EM} &= \frac{MF + PM}{MF} \quad (\text{since } PM = MQ) \\ \Rightarrow \frac{MQ}{EM} &= \frac{PM}{MF} \\ \Rightarrow EM &= MF. \end{aligned}$$

□

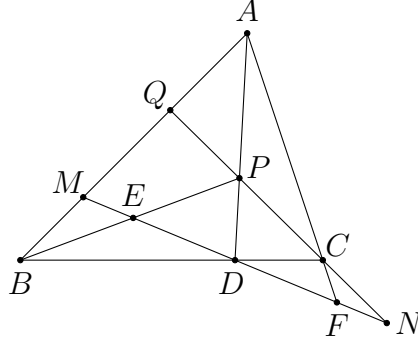
Remarks. Here, one may or may not use directed lengths since the configuration is unique.

Note that the butterfly theorem is also valid when PQ lies outside the circle, in which case M is the point on the line such that $OM \perp PQ$. However, the method used in this solution does not seem to work in the general case.

Problem 1.2. (CSMO 2004 Problem 2) Let D be a point on the side BC of $\triangle ABC$. A point P lies on the segment AD . A line through D intersects the segments AB, PB at points M, E respectively, and intersects the extensions of the segments AC, PC at points F, N respectively. If $DE = DF$, prove that $DM = DN$.

Analysis. The situation in this problem is quite similar to [problem 1.1](#), especially for the line MN . However, we need to add an extra point this time to carry out the projectivity. When we use projective geometry, usually we define some new points in a natural way, as they are mostly the intersections of existing lines. For example, we start from the pencil $MN(M, D, F, N)$ in the solution and try to project D to B . So we label the intersection of AB and PC as shown.

Solution.



Let $Q = AB \cap PC$. Then we have the projectivity

$$MN(M, D, F, N) \xrightarrow{C} AB(M, B, A, Q) \xrightarrow{P} MN(M, E, D, N).$$

Therefore, we get $(M, D; F, N) = (M, E; D, N)$. Now,

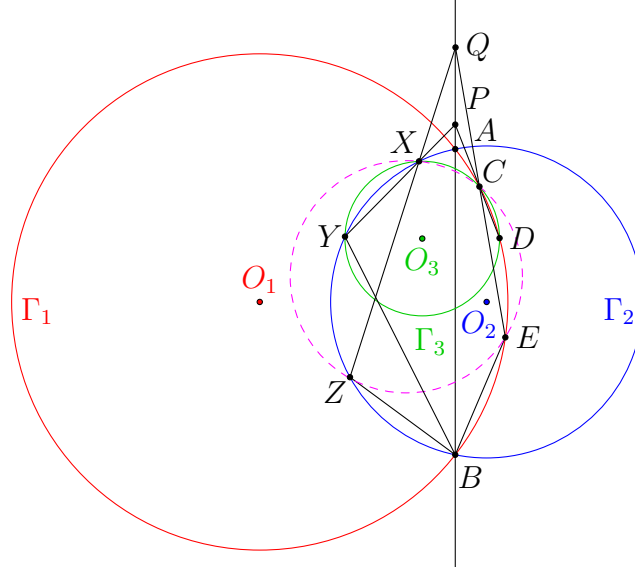
$$\begin{aligned} (M, D; F, N) &= (M, E; D, N) \\ \Rightarrow \frac{\frac{MF}{MN}}{\frac{DF}{DN}} &= \frac{\frac{ME}{MN}}{\frac{ED}{EN}} \\ \Rightarrow \frac{DN(MD + DF)}{MN \cdot DF} &= \frac{MD(ED + DN)}{MN \cdot ED} \\ \Rightarrow DN(MD + DF) &= MD(ED + DN) \quad (\text{since } DF = ED) \\ \Rightarrow DN \cdot DF &= MD \cdot ED \\ \Rightarrow DN &= MD. \end{aligned}$$

□

Problem 1.3. Two circles Γ_1 and Γ_2 with centres O_1 and O_2 intersect at points A and B . Another circle Γ_3 with centre lying inside the overlapping region of Γ_1 and Γ_2 meets Γ_1 at C, D and meets Γ_2 at X, Y respectively. Let E, Z be the reflections of A in the lines DO_1, YO_2 respectively. Suppose A, C, D, E, B and A, X, Y, Z, B lie in these orders on the two circles. Prove that E, C, X, Z are concyclic iff $BE = BZ$.

Analysis. For problems involving more than one circle, we usually consider the radical axes of some pairs of circles. Therefore, in our first step, we shall construct intersection points of some lines to the radical axis of the two large circles. To get rid of the length BE , we need to consider the cross ratio of some points on Γ_1 . Then we may project the points onto the radical axis AB for comparison.

Solution.



Let P be the radical centre of $\Gamma_1, \Gamma_2, \Gamma_3$ (i.e. the intersection point of AB, CD, XY). Let AB meet CE, XZ at Q_1, Q_2 respectively. We have

$$AB(Q_1, A, P, B) \xrightarrow{C} \Gamma_1(E, A, D, B)$$

and similarly,

$$AB(Q_2, A, P, B) \xrightarrow{X} \Gamma_2(Z, A, Y, B).$$

So $(Q_1, A; P, B) = (E, A; D, B)$ and $(Q_2, A; P, B) = (Z, A; Y, B)$.

By the well-known fact of radical centre, we know that E, C, X, Z are concyclic iff AB, CE, XZ are concurrent, i.e. $Q_1 = Q_2$.

We now consider the magnitudes of $(E, A; D, B)$ and $(Z, A; Y, B)$ (for lengths of chords on circles, we do not have directions).

$$\begin{aligned} \frac{ED}{EB} &= \frac{ZY}{ZB} \\ \frac{AD}{AB} &= \frac{AY}{AB} \\ \Leftrightarrow \frac{AB}{EB} &= \frac{AB}{ZB} \quad (\text{as } AD = ED \text{ and } AY = ZY) \\ \Leftrightarrow EB &= ZB. \end{aligned}$$

From the configuration, it is clear that both $(E, A; D, B)$ and $(Z, A; Y, B)$ are negative. So $Q_1 = Q_2$ iff $(E, A; D, B) = (Z, A; Y, B)$ iff $EB = ZB$. $\square$

Remarks. Although the figure is difficult to draw, one does not really need it using projective geometry!

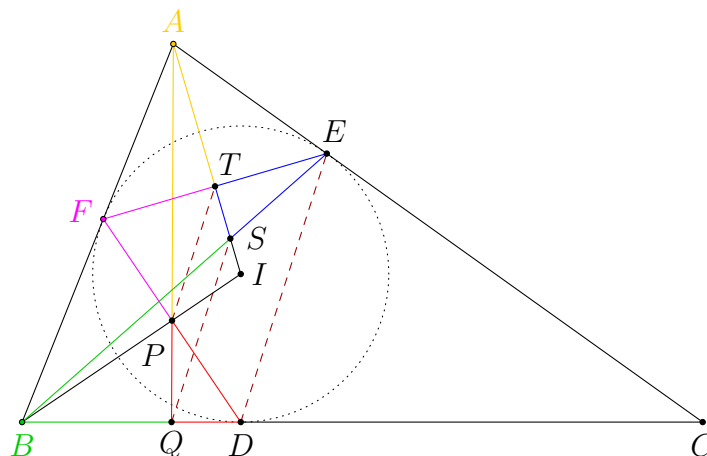
Next, we go through the applications of the projective theorems introduced.

Problem 1.4. Let I be the incentre of $\triangle ABC$. The incircle touches the sides BC, CA, AB at D, E, F respectively. Suppose BI meets FD at P , AP meets BC at Q , and BE meets AI at S . Prove that QS and DE are parallel.

Analysis. It is not hard to figure out a proof for this problem. In this solution, we shall demonstrate the use of Desargues' theorem. This theorem can be used to prove concurrent lines or collinear points, and is particularly useful when there is no special condition on the triangle.

To prove $QS \parallel DE$ in this problem, it means the two lines intersect at a point at infinity. So we may find another line which is parallel to them, for example, the line joining the midpoints of FD and FE .

Solution.



Let $T = AI \cap EF$. For $\triangle QDP$ and $\triangle SET$, as QD, DP, PQ meet SE, ET, TS at the collinear points B, F, A respectively, the two triangles are perspective from a point by Desargues' theorem. This means QS, DE, PT are concurrent in the projective sense. Note that P, T are the midpoints of FD, FE respectively. So $DE \parallel PT$ by the midpoint theorem, and hence $QS \parallel DE$. $\square$

Remarks. It is rare that a direct application of Desargues' theorem (and other theorems as well) could solve the whole problem, except for this kind of trivial problems. However, sometimes applying the theorem to get extra information is helpful. Therefore, we may take some pairs of triangles from the figure for testing whenever we have nowhere to start a problem. If some pairs are perspective to each other, more information can be obtained. Of course sometimes we need to include additional points for consideration.

Problem 1.5. (IMO Shortlist 2006 G2) Let $ABCD$ be a trapezoid with parallel sides $AB > CD$. Points K and L lie on the line segments AB and CD , respectively,

so that $\frac{AK}{KB} = \frac{DL}{LC}$. Suppose that there are points P and Q on the line segment KL satisfying

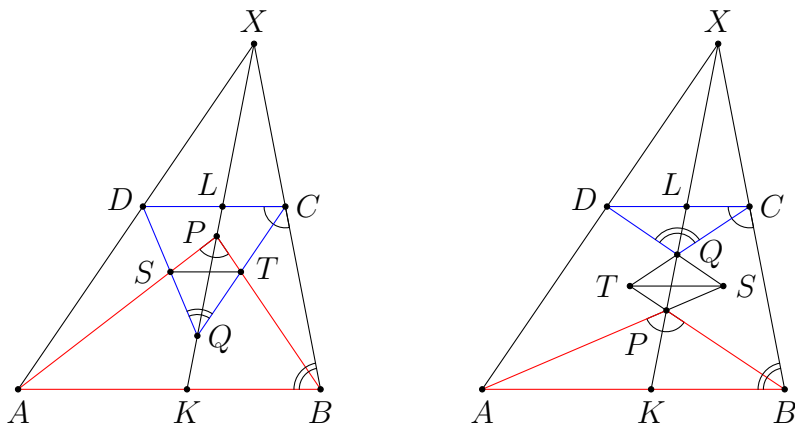
$$\angle APB = \angle BCD \quad \text{and} \quad \angle CQD = \angle ABC.$$

Prove that the points P, Q, B and C are concyclic.

Analysis. From the configuration, it seems that the easiest way to establish the concyclicity is to prove $\sphericalangle CQP = \sphericalangle CBP$. By angle chasing, it suffices to prove that some two lines are parallel, which can be done by Desargues' theorem as in the previous problem.

Note that the condition $\frac{AK}{KB} = \frac{DL}{LC}$ is equivalent to AD, KL, BC being concurrent by the intercept theorem, and so this gives us a way to find two perspective triangles.

Solution.



Let $S = AP \cap DQ$ and $T = BP \cap CQ$. As $AB \parallel DC$ and $\frac{AK}{KB} = \frac{DL}{LC}$, the lines AD, KL, BC are concurrent at some point X by the intercept theorem. In particular, $\triangle APB$ and $\triangle DQC$ are perspective from the point X . By Desargues' theorem, they are perspective from a line. Note that $S = AP \cap DQ$, $T = PB \cap QC$ and the remaining intersection point $BA \cap CD$ is a point at infinity. This means $BA \parallel CD \parallel TS$.

To deal with the two configurations together, we use directed angles. From

$$\angle SPT = \angle APB = \angle DCB = \angle ABC = \angle DQC = \angle SQT,$$

points P, S, Q, T are concyclic. Then

$$\angle CQP = \angle CQS + \angle SQP = \angle CQD + \angle STP = \angle CBA + \angle ABP = \angle CBP,$$

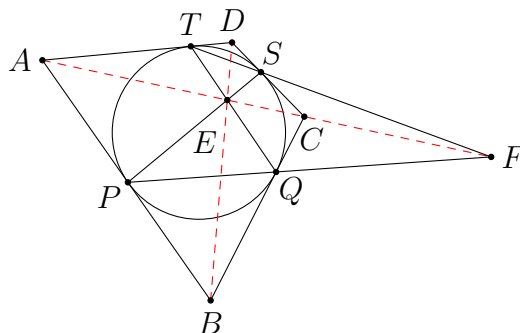
which implies P, Q, B, C are concyclic.

The next few problems are applications of Pascal's theorem. This theorem is useful when there are many points on a circle.

Problem 1.6. Let $ABCD$ be a circumscribed quadrilateral. Its incircle touches the sides AB, BC, CD, DA at P, Q, S, T respectively. Prove that AC, BD, PS, QT are concurrent, and AC, PQ, TS are concurrent.

Analysis. Although there are only four points on the circle, their tangents are involved, and so it is still possible to apply Pascal's theorem. For example, A is the intersection of PP and TT . So we may consider $PPXTTY$ for some points X, Y . Indeed, there is only one choice up to permutation if no additional point is added.

Solution.



Consider $PPQTTS$. By Pascal's theorem, points $A = PP \cap TT$, $F = PQ \cap TS$ and $E = QT \cap SP$ are collinear. Due to symmetry, we deduce C, E, F are collinear by applying Pascal's theorem to $SSPQQT$. This means AC, PQ, TS are concurrent at F , while AC, PS, QT are concurrent at E . Note that BD also passes through the concurrent point of the latter by symmetry. The proof is thus complete.

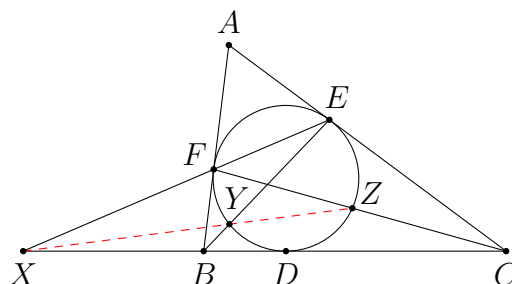
Alternatively, one may simply use Brianchon's theorem in the degenerate hexagon $BQCDDTA$ since the incircle is tangent to the sides BQ, QC, CD, DT, TA, AB . This immediately yields BD, QT, CA are concurrent, similarly for PS . $\square$

Remarks. When we apply Pascal's theorem, there is no need to plot the exact intersection points in the figure. We only need to consider six points 123456 in order and conclude that $12 \cap 45$, $23 \cap 56$ and $34 \cap 61$ are collinear. This can be viewed as an advantage for those who cannot draw an accurate figure.

Problem 1.7. (CHKMO 2011 Problem 2) The incircle of a triangle ABC touches the three sides BC, CA and AB at D, E and F respectively. The line segments BE and CF intersect the incircle at Y and Z respectively. Suppose the lines FE and BC intersect externally at X . Prove that the three points X, Y and Z are collinear.

Analysis. This time there are five points on the circle. However, for point D , there is only a tangent but not other lines passing through it. Without adding extra lines, the simplest way is to consider only points E, F, Y, Z . The deduction is the same as the previous problem.

Solution.



Let $X' = EF \cap YZ$. Apply Pascal's theorem to $EEFFZY$. Then $C = EE \cap FZ$, $X' = EF \cap ZY$ and $B = FF \cap YE$ are collinear. Therefore, $X' = X$, and X, Y, Z are collinear. $\square$

Remarks. One may apply Pascal's theorem to a group of points involving D to derive additional information. Such facts may be useful to solve other related problems.

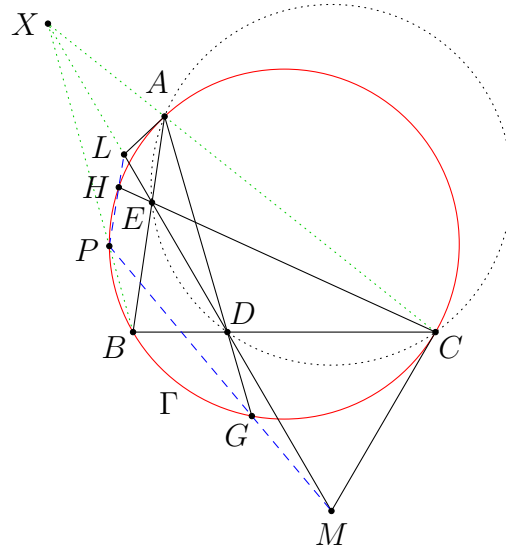
Problem 1.8. (APMO 2008 Problem 3) Let Γ be the circumcircle of a triangle ABC . A circle passing through points A and C meets the sides BC and BA at D and E , respectively. The lines AD and CE meet Γ again at G and H , respectively. The tangent lines of Γ at A and C meet the line DE at L and M , respectively. Prove that the lines LH and MG meet at Γ .

Analysis. Pascal's theorem is useful for this kind of problems, where only intersections of lines joining two points on a circle and tangents are involved, and there is no special property of the triangle. We only need to think of how to use this theorem.

Note that the conclusion of Pascal's theorem is that three points not lying on the circle are collinear. This is very different from what we are going to prove. Even if we first mark the second intersection point of LH and Γ , and attempt to prove that it lies on MG , this cannot be derived directly by Pascal's theorem. Therefore, we may reconstruct the figure and try to prove an equivalent statement where three points not on Γ are collinear. One good choice for the line on which the points lie is DE , since there are many points on that line, none of which lies on Γ .

Solution. Let P be the second intersection of MG and Γ , and let L' be the intersection of PH and the tangent at A . It suffices to prove L' lies on DE .

To generate the intersection point L' , the hexagon in Pascal's theorem must be $PH * AA*$. After some trial, we apply Pascal's theorem to $PHCAAB$. This implies $L' = PH \cap AA$, $E = HC \cap AB$ and $X = CA \cap BP$ (say) are collinear. Symmetrically, we apply Pascal's theorem to $PGACCB$. Then $M = PG \cap CC$, $D = GA \cap CB$ and $X = AC \cap BP$ are collinear. Therefore, all the points L', E, D, M, X lie on the same line. In particular, we have $L' = DE \cap AA = L$, and the conclusion follows. $\square$



Remarks. The condition that A, C, D, E are concyclic is redundant!

Problem 1.9. (IMO Shortlist 2004 G2) Let Γ be a circle and let d be a line such that Γ and d have no common points. Further, let AB be a diameter of the circle Γ ; assume that this diameter AB is perpendicular to the line d , and the point B is nearer to the line d than the point A . Let C be an arbitrary point on the circle Γ , different from the points A and B . Let D be the point of intersection of the lines AC and d . One of the two tangents from the point D to the circle Γ touches this circle Γ at a point E ; hereby, we assume that the points B and E lie in the same halfplane with respect to the line AC . Denote by F the point of intersection of the lines BE and d . Let the line AF intersect the circle Γ at a point G , different from A .

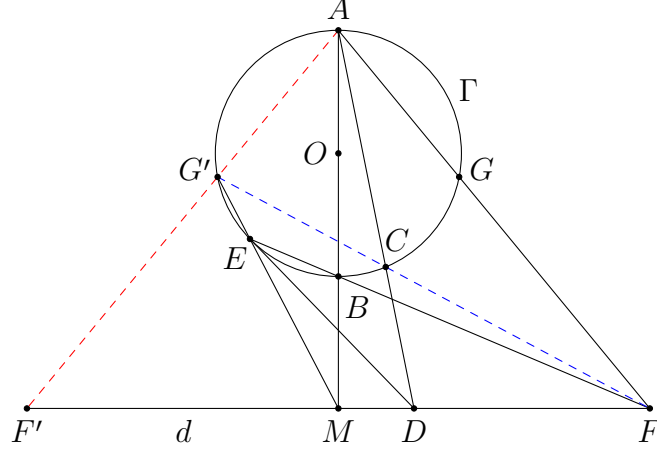
Prove that the reflection of the point G in the line AB lies on the line CF .

Analysis. The configuration is quite similar to that in the general case of the butterfly theorem. To apply this theorem, we should locate the intersection of AB and d , and find a pair of points on d whose midpoint is that intersection point. Since there are quite a lot of points on the circle, it is natural to use Pascal's theorem as well.

Solution. Let F' be the reflection of the point F in the line AB , and let $M = AB \cap d$ and $G' = ME \cap \Gamma$. We apply the butterfly theorem to d and secants MBA and MEG' . This shows $AG' \cap d$ and $BE \cap d$ are symmetric in the point M . Therefore, A, G', F' are collinear, and hence G' is the reflection of G in the line AB .

Next, we would like to use Pascal's theorem. Clearly, we would like the three intersection points to lie on d . To use the tangent at E , one of the intersection points must be $D = EE \cap CA$. It is not hard to fill up the remaining two entries. Indeed, we use Pascal's theorem in $EEG' CAB$. This shows the points $D = EE \cap CA$, $M = EG' \cap AB$

and $G'C \cap BE$ are collinear, all lying on d . Equivalently, $G'C, BE, d$ are concurrent. Therefore, $F = G'C \cap BE$, and hence G', C, F are collinear. $\square$



Problem 1.10. (CMO 2005 Problem 2) A circle intersects the sides BC, CA, AB of $\triangle ABC$ at $D_1, D_2; E_1, E_2; F_1, F_2$ in this order. The segments D_1E_1 and D_2F_2 intersect at a point L . The segments E_1F_1 and E_2D_2 intersect at a point M . The segments F_1D_1 and F_2E_2 intersect at a point N . Prove that AL, BM, CN are concurrent.

Analysis. We shall give an application of projective pencils of lines.

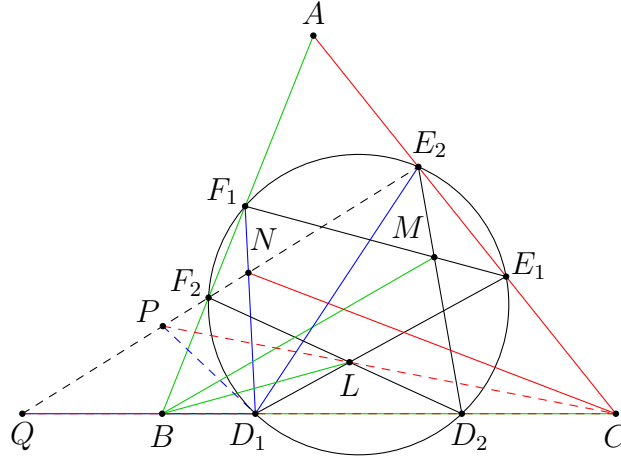
We may rephrase the conclusion as a result about perspective pencils. For example, if we have two projective pencils $B(\ell_1, \ell_2, \ell_3, BC)$ and $C(m_1, m_2, m_3, BC)$, then we know that $\ell_1 \cap m_1, \ell_2 \cap m_2$ and $\ell_3 \cap m_3$ are collinear by [proposition 1.2](#). Since AL does not pass through B or C , an appropriate choice is that A and L are two of the intersection points. The remaining intersection point should be $BM \cap CN$. Thus, we are going to show $B(BA, BM, BL, BC)$ and $C(CA, CN, CL, CB)$ are projective.

To get rid of the point N , WLOG we may consider the line E_2F_2 . But then we need more points on this line to perform projections. In particular, it is fruitful if we have some information on the intersection point of E_2F_2 and CL . This is why we shall apply Pascal's theorem in the first step. There should be no doubt on the choice of the 6 points, since C and L are determined only by the intersections of two secants of the circle.

Solution. Let $P = E_2F_2 \cap D_1D_1$ and $Q = E_2F_2 \cap CB$. Apply Pascal's theorem to points $F_2D_2D_1D_1E_1E_2$. The points $L = F_2D_2 \cap D_1E_1$, $C = D_2D_1 \cap E_1E_2$ and $P = D_1D_1 \cap E_2F_2$ are collinear. We project the points as follows.

$$\begin{aligned} C(CA, CN, CL, CB) &\xrightarrow{E_2F_2} D_1(D_1E_2, D_1N, D_1P, D_1Q) \\ &= D_1(D_1E_2, D_1F_1, D_1D_1, D_1D_2) \end{aligned}$$

Symmetrically, $B(BA, BM, BL, BC)$ and $D_2(D_2F_1, D_2E_2, D_2D_2, D_2D_1)$ are projective to each other. Note that the latter is projective to $D_2(D_2E_2, D_2F_1, D_2D_1, D_2D_2)$ as they have the same cross ratio.



Next, D_1, D_2, E_2, F_1 lie on a conic. Then we can use a projection on conics to establish

$$D_2(D_2E_2, D_2F_1, D_2D_1, D_2D_2) \longrightarrow D_1(D_1E_2, D_1F_1, D_1D_1, D_1D_2).$$

Combining all these, we know that $C(CA, CN, CL, CB)$ and $B(BA, BM, BL, BC)$ are projective. Since BC is an invariant, [proposition 1.2](#) shows that this is a perspectivity, and hence $A = CA \cap BA$, $CN \cap BM$ and $L = CL \cap BL$ are collinear. In other words, AL, BM, CN are concurrent. $\square$

Problem 1.11. (USAMO 2005 Problem 3) Let ABC be an acute-angled triangle, and let P and Q be two points on its side BC . Construct a point C_1 in such a way that the convex quadrilateral $APBC_1$ is cyclic, $QC_1 \parallel CA$, and C_1 and Q lie on opposite sides of line AB . Construct a point B_1 in such a way that the convex quadrilateral $APCB_1$ is cyclic, $QB_1 \parallel BA$, and B_1 and Q lie on opposite sides of line AC . Prove that the points B_1, C_1, P , and Q lie on a circle.

Analysis. If one tries to draw the figure, one may observe that C_1, A, B_1 are collinear. The result follows readily once this is obtained. We shall show the usage of Pappus' theorem in the solution below.

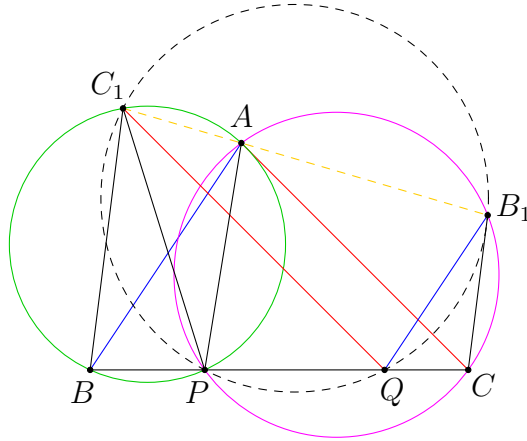
Solution. Let B'_1 be the second intersection of C_1A and (APC) . We are going to show that $B'_1 = B_1$. Note that

$$\angle PBC_1 = \angle PAB'_1 = 180^\circ - \angle B'_1CP,$$

which implies $BC_1 \parallel CB'_1$. Apply Pappus' theorem to points B, Q, C and B'_1, A, C_1 . This shows the points $BA \cap QB'_1$, $BC_1 \cap CB'_1$ and $QC_1 \cap CA$ are collinear. Since $BC_1 \parallel CB'_1$ and $QC_1 \parallel CA$, two of the intersection points are points at infinity. Thus, the line of Pappus is the line at infinity. Then $BA \parallel QB'_1$ and $B'_1 = B_1$, and hence C_1, A, B_1 are collinear. It is easy to get

$$\angle PC_1B_1 = \angle PBA = \angle CQB_1.$$

This implies B_1, C_1, P, Q are concyclic. $\square$

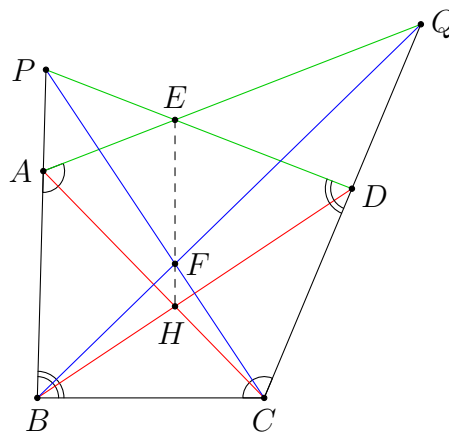


Remarks. Pappus' theorem occurs less frequently than Pascal's theorem. But it is still useful in some occasions.

Problem 1.12. (IMO Shortlist 2017 G1) Let $ABCDE$ be a convex pentagon such that $AB = BC = CD$, $\angle EAB = \angle BCD$, and $\angle EDC = \angle CBA$. Prove that the perpendicular line from E to BC and the line segments AC and BD are concurrent.

Analysis. As $BC = CD$, we have $\angle CBD = \angle CDB$. This gives $\angle DBA = \angle EDB$. Thus, one way to use these conditions is to construct the intersection point of BA and DE , so that an isosceles triangle is formed. We can do the same on the other symmetric side. Afterwards, the figure looks like the one in Pappus' theorem.

Solution.



Let AB meet DE at P , and let CD meet AE at Q . Since $BC = CD$, we have $\angle CBD = \angle CDB$. So

$$\angle DBA = \angle CBA - \angle CBD = \angle EDC - \angle BDC = \angle EDB.$$

This implies $\triangle PBD$ is isosceles. Together with $BC = CD$, we find that PC is the perpendicular bisector of BD . Similarly, QB is the perpendicular bisector of AC .

Apply Pappus' theorem to the lines BAP and CDQ . Then the points $H = BD \cap AC$, $F = BQ \cap PC$ and $E = AQ \cap PD$ are collinear. From above, we have $FC \perp BH$ and $FB \perp CH$. So H is the orthocentre of $\triangle FBC$. This implies $FH \perp BC$. It follows that the perpendicular line from E to BC is FH , which passes through the intersection point H of AC and BD as desired. $\square$

Problem 1.13. (IMO Shortlist 2019 G3/Problem 2) In triangle ABC , point A_1 lies on side BC and point B_1 lies on side AC . Let P and Q be points on segments AA_1 and BB_1 , respectively, such that PQ is parallel to AB . Let P_1 be a point on line PB_1 , such that B_1 lies strictly between P and P_1 , and $\angle PP_1C = \angle BAC$. Similarly, let Q_1 be a point on line QA_1 , such that A_1 lies strictly between Q and Q_1 , and $\angle CQ_1Q = \angle CBA$.

Prove that points P, Q, P_1 , and Q_1 are concyclic.

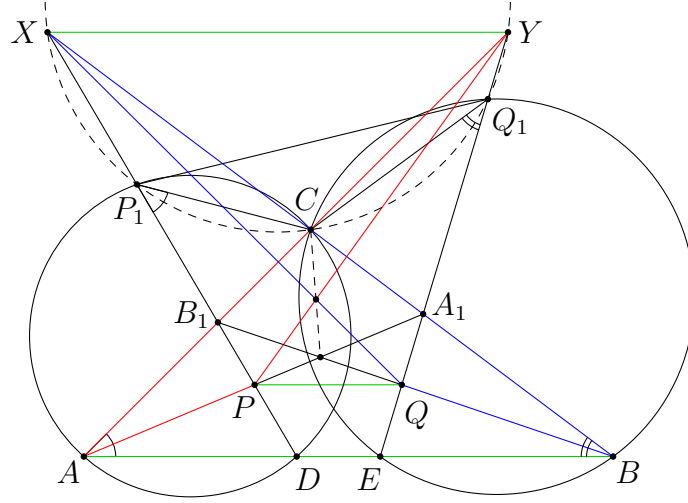
Analysis. The first idea is to handle the angle conditions. As the pair of equal angles looks like angles in the same segment, an obvious way to use this is to define the intersection point D of B_1P and AB , so that P_1, A, D, C are concyclic. This is indeed the easiest way to construct P_1 .

The second idea comes from the following fact (which was a problem that we did in the training in year 2018–19, though no one could recall this fact in the IMO). The points are labelled so as to fit our problem.

Let X and Y be points on the sidelines BC and AC of $\triangle ABC$ respectively such that $XY \parallel AB$. Let D and E be arbitrary points on the line AB . The circumcircle of $\triangle ACD$ meets the line DX again at P_1 . The circumcircle of $\triangle BCE$ meets the line EY again at Q_1 . Then X, P_1, C, Q_1, Y are concyclic. (Equivalently, when D travels along AB , the locus of P_1 is the circumcircle of $\triangle CXY$. The same holds for E and Q_1 by symmetry.)

In view of this result, we construct the points $X = B_1P \cap CB$ and $Y = A_1Q \cap CA$, and attempt to prove $XY \parallel AB$. Once this is proved, the remaining steps are just simple angle chasing (if one knows the concept of antiparallelism, this is even more obvious as it simply means P_1Q_1 is antiparallel to XY , and hence to PQ). As we should have a group of parallel lines XY, PQ, AB , we may apply Desargues' theorem. There are several possible choices for the pair of triangles, such as $\triangle XPA$ and $\triangle YQB$, or $\triangle XQA$ and $\triangle YPB$, etc. We just need to see which result is the easiest to prove.

Solution. Let $D = B_1P \cap AB$, $E = A_1Q \cap AB$, $X = B_1P \cap CB$ and $Y = A_1Q \cap CA$. From $\angle PP_1C = \angle BAC$, we know that P_1, A, D, C are concyclic. Similarly, Q_1, B, E, C are concyclic.



Applying Pappus' theorem to the lines XB_1P and YA_1Q , we find that the points $C = XA_1 \cap YB_1$, $XQ \cap YP$ and $B_1Q \cap A_1P$ are collinear. This shows $\triangle APY$ and $\triangle BQX$ are perspective from a line, since these are exactly the three intersection points of the corresponding sides. By Desargues' theorem, the lines XY, PQ, AB are concurrent. Since $PQ \parallel AB$, this means XY is parallel to PQ and AB . It follows that

$$\angle CP_1X = \angle CP_1D = \angle CAD = \angle CYX.$$

So C, Y, X, P_1 are concyclic. By symmetry, Q_1 also lies on this circle. Thus,

$$\angle QPP_1 = \angle YXP_1 = \angle QQ_1P_1.$$

This implies P, Q, Q_1, P_1 are concyclic. □

Remarks. It is a pity that only 6 points instead of 7 will be awarded to this solution according to the marking scheme. This is because the proof fails when X and Y are points at infinity. The steps involving projective geometry are fine. Indeed, we can prove P_1, C, Q_1 are collinear in that case. But the part about angle chasing should be modified. One may try to fill in the details.

Here is the summary of the tools that we have learnt in this section.

- perspective transformation ([problems 1.1, 1.2, 1.3, 1.10](#))
- cross ratio ([problems 1.1, 1.2, 1.3](#))
- Desargues' theorem ([problems 1.4, 1.5, 1.13](#))
- Pascal's theorem ([problems 1.6, 1.7, 1.8, 1.9, 1.10](#))
- Pappus' theorem ([problems 1.11, 1.12, 1.13](#))

2 Harmonic Division

In this section, we introduce a useful concept in projective geometry. It is about a special pencil of four points, which naturally occurs in many geometry problems. Some of the properties of this kind of pencils are closely related to different ingredients in Euclidean geometry, such as midpoints, angles, tangents etc.

We first give the definition. A pencil $\ell(A, B, C, D)$ of four distinct points on a line ℓ is called *harmonic* if $(A, B; C, D) = -1$. In that case, we say that $A, B; C, D$ form a *harmonic division*(調和分割), or A, B is a *harmonic conjugate pair*(調和共軛) w.r.t. CD . Similarly, a pencil $\Gamma(A, B, C, D)$ on a conic Γ is *harmonic* if $(A, B; C, D) = -1$.

Since this concept is defined by cross ratios, it is obvious that the order of points is important. By definition, $(A, B; C, D) = -1$ iff any of

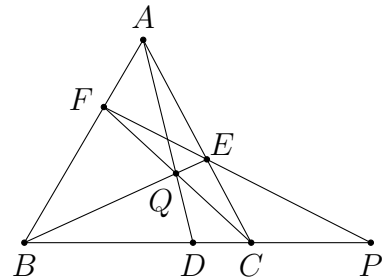
$$(A, B; D, C), (B, A; C, D), (B, A; D, C), (C, D; A, B), \\ (C, D; B, A), (D, C; A, B), (D, C; B, A)$$

is -1 . In other words, whether $A, B; C, D$ form a harmonic division or not is an intrinsic property of the two pairs of points on a line. This suggests why we say that A, B and C, D are harmonic conjugates. To have a negative cross ratio, the pairs A, B and C, D must be separated by each other. For example, the order can be A, C, B, D .

Next, we shall introduce some common harmonic divisions.

Proposition 2.1. In $\triangle ABC$, the points D, E, F lie on the sidelines BC, CA, AB respectively. Let P be another point on the line BC . Then any two of the following statements imply the third one.

- (i) $(B, C; D, P) = -1$
- (ii) AD, BE, CF are concurrent
- (iii) P, E, F are collinear



Proof. Suppose we have (i). This means $\frac{BD}{DC} = -\frac{BP}{CP}$. The relation can be rewritten

as $\frac{BD}{DC} = -\frac{BP}{PC}$. By Ceva's theorem and Menelaus' theorem, (ii) holds iff

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1, \tag{1}$$

while (iii) holds iff

$$\frac{BP}{PC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = -1. \quad (2)$$

These are equivalent from the given condition.

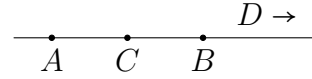
Next, suppose we have (ii) and (iii). As above, we have (1) and (2). By comparing these equations, we find that $\frac{BD}{DC} = -\frac{BP}{PC}$, i.e. $(B, C; D, P) = -1$, and (i) follows. $\square$

The configuration in proposition 2.1 only involves some concurrent lines and collinear points, which is rather general. It should appear in many geometric problems and could be applied to show concurrency and collinearity. In fact, this is considered as the definition of harmonic division in projective geometry, since we do not have any quantitative meaning of lengths and cross ratios in projective geometry.

However, this fact is not as important as those introduced below, since this still lies within the scope of projective geometry and does not have many connections with Euclidean geometry. Also, the things introduced in section 1 are sufficient to give the same results.

Proposition 2.2. Let A, B, C, D be distinct points on a line ℓ in any order. Then any two of the following statements imply the third one.

- (i) $(A, B; C, D) = -1$
- (ii) C is the midpoint of AB
- (iii) D is the ideal point of ℓ



Proof. The proof is again based on the definition. Suppose we have (i). It is easy to deduce $-\frac{AC}{BC} = \frac{AD}{BD}$. Noting (ii) holds iff $-\frac{AC}{BC} = 1$, while (iii) holds iff $\frac{AD}{BD} = 1$, these are equivalent from the condition.

Suppose we have (ii) and (iii). Then

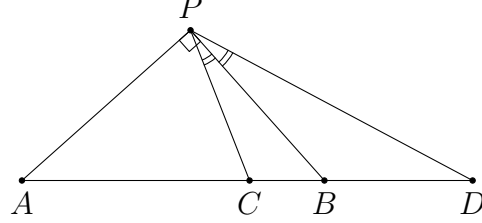
$$(A, B; C, D) = \frac{\frac{AC}{BC}}{\frac{AD}{BD}} = \frac{\frac{AC}{BC}}{\frac{AD}{BD}} = \frac{-1}{1} = -1,$$

and (i) holds. $\square$

This limiting case is quite useful as we often come across midpoints in Olympiad problems. The point at infinity can be easily converted to ordinary points by considering projections through parallel lines.

Proposition 2.3. Let A, C, B, D be distinct points on a line in this order, and let P be a point not on the line. Then any two of the following statements imply the third one.

- (i) $(A, B; C, D) = -1$
- (ii) $\angle APB = 90^\circ$
- (iii) $\angle CPB = \angle BPD$



Proof. By the ratio lemma, we have

$$\frac{AC}{AD} = \frac{PC \sin \angle APC}{PD \sin \angle APD}$$

and

$$\frac{BC}{BD} = \frac{PC \sin \angle BPC}{PD \sin \angle BPD}.$$

This yields

$$\frac{\frac{AC}{AD}}{\frac{BC}{BD}} = \frac{\sin \angle APC}{\sin \angle APD} \cdot \frac{\sin \angle BPD}{\sin \angle BPC}. \quad (1)$$

Suppose we have (iii). Let $\angle CPB = \angle BPD = \theta \neq 0^\circ$. From the relative positions, we know that (i) holds iff $\frac{\frac{AC}{AD}}{\frac{BC}{BD}} = 1$ (in Euclidean length) iff $\sin \angle APC = \sin \angle APD$ from (1). This is equivalent to

$$\sin(\angle APB - \theta) = \sin(\angle APB + \theta).$$

As $\theta \neq 0^\circ$, this is true iff $(\angle APB - \theta) + (\angle APB + \theta) = 180^\circ$ iff (ii) holds.

Next, suppose we have (i) and (ii). Then

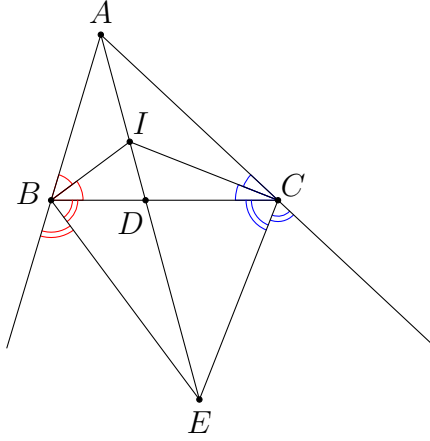
$$\begin{aligned} & \frac{\sin \angle APC}{\sin \angle APD} = \frac{\sin \angle BPC}{\sin \angle BPD} \\ \Rightarrow & \frac{\cos \angle BPC}{\cos \angle BPD} = \frac{\sin \angle BPC}{\sin \angle BPD} \\ \Rightarrow & \tan \angle BPD = \tan \angle BPC \\ \Rightarrow & \angle BPD = \angle BPC. \end{aligned}$$

This yields (iii). □

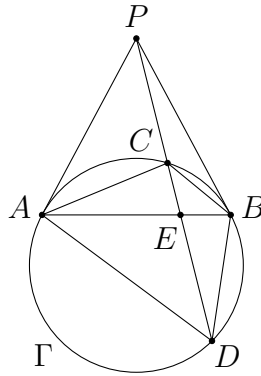
Proposition 2.3 involves angles, which is the main ingredient in Euclidean geometry. In particular, right angles and angle bisectors occur frequently in geometry problems.

To use this proposition, sometimes we need to locate additional points to move the equal angles and right angles together. Conditions (ii) and (iii) mean that A and B lie on the external and internal angle bisectors of $\angle CPD$ respectively. This leads to the following standard example.

Let I and E be the incentre and the A -excentre of $\triangle ABC$, and let D be the foot of internal angle bisector of $\angle BAC$. Since $\angle EBI = 90^\circ$ and $\angle IBA = \angle DBI$, we have $(A, D; I, E) = -1$ by [proposition 2.3](#).



Proposition 2.4. Let A, B, C, D be points on a circle Γ . Then $(A, B; C, D) = -1$ iff the tangent at A , the tangent at B and the line CD are concurrent. In that case, let $P = AA \cap BB$ and let E be any point on the line CD . Then $(P, E; C, D) = -1$ iff $E = AB \cap CD$.



Proof. Let $P = AA \cap BB$. Suppose P, C, D are collinear. From $\triangle PAC \sim \triangle PDA$, we have

$$\frac{AC}{DA} = \frac{PA}{PD} = \frac{PC}{PA} = \sqrt{\frac{PA}{PD} \cdot \frac{PC}{PA}} = \sqrt{\frac{PC}{PD}}.$$

Similarly, $\Delta PBC \sim \Delta PDB$ implies

$$\frac{BC}{DB} = \sqrt{\frac{PC}{PD}} = \frac{AC}{DA}.$$

This gives $\frac{AC}{AD} = \frac{BC}{BD}$ in Euclidean sense. Note that A, B must be separated by C, D on the circle. Thus, a negative sign should be added to the cross ratio and we get $(A, B; C, D) = -1$.

Conversely, suppose $(A, B; C, D) = -1$. Let PC intersect Γ for the second time at D' . Then $(A, B; C, D') = -1$ from above, and hence $D = D'$. This yields P, C, D are collinear. Note that the result is also true when P is a point at infinity. In that case we do not need to argue using similar triangles since we have $AC = AD$ and $BC = BD$.

For the second statement, consider $E' = AB \cap CD$. We have the projection

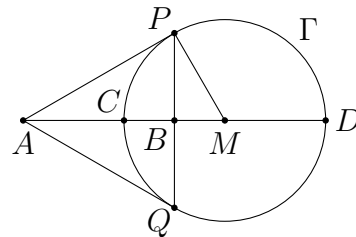
$$\Gamma(A, B, C, D) \xrightarrow{A} CD(P, E', C, D).$$

This implies $(P, E'; C, D) = -1$. Thus, we find that $(P, E; C, D) = -1$ iff $E = E'$ iff $E = AB \cap CD$. $\square$

This is the most important fact related to a harmonic division on a circle. Note that this is also related to a well-known fact about *symmedians*(類似中線). Recall that the symmedian from C of $\triangle ABC$ is the reflection of the C -median w.r.t. the internal angle bisector from C . This symmedian is concurrent with the two tangent lines at A and B to the circumcircle. So $A, B; C, D$ form a harmonic division on a circle iff CD is the symmedian from C of $\triangle ABC$.

Proposition 2.5. Let A, C, B, D be distinct points on a line in this order, and let M be the midpoint of CD . Then the following are equivalent (in directed lengths).

- (i) $(A, B; C, D) = -1$
- (ii) $AC \cdot AD = AB \cdot AM$
- (iii) $MA \cdot MB = MC^2$



Proof. Let Γ be the circle with CD as a diameter. So M is the centre of Γ . Let AP and AQ be the tangents from A to Γ with contact points P and Q respectively. Then $AC \cdot AD = AP^2$ by the power of a point theorem.

Suppose we have (i). By [proposition 2.4](#), P, B, Q are collinear, and so $PB \perp AM$. Then $\triangle ABP \sim \triangle APM$. In particular, $\frac{AB}{AP} = \frac{AP}{AM}$, and hence

$$AC \cdot AD = AP^2 = AB \cdot AM.$$

Similarly, $\triangle MBP \sim \triangle MPA$ yields $\frac{MB}{MP} = \frac{MP}{MA}$. Then

$$MA \cdot MB = MP^2 = MC^2.$$

From the configuration, the signs of the expressions in (ii) and (iii) are valid. Both converses can be derived easily by the method of false position. $\square$

This fact is as well important since the expressions (ii) and (iii) are closely related to the expressions for power of points. It can be used to prove concyclicity. Note that the converses are valid only when directed lengths are used. So one should check the signs or the configuration carefully.

To use the concept of harmonic division, one way is to establish a projection between two groups of harmonic pencils directly from the well-known configurations, and then use the skills introduced in [section 1](#) to deduce more facts. Another way is to project a harmonic pencil to some particular points, hence finding a simple relationship among the lengths of the segments on that pencil or some special facts as introduced. This might sometimes simplify a series of application of Ceva's theorem, Menelaus' theorem and/or sine law to some obvious projectivities.

It is always simpler to learn by going through various problems.

Problem 2.1. (CWMO 2005 Problem 2) Two tangents PA, PB are drawn from a point P outside a circle where A, B are contact points. Another secant to the circle through P intersects the circle at the points C, D . A line through B parallel to PA intersects the lines AC, AD at E, F respectively. Prove that $BE = BF$.

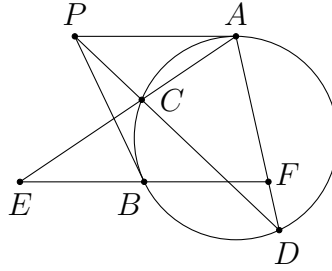
Analysis. We first give a simple example to illustrate a direct use of harmonic division. The reason why we might think of this method is that we immediately obtain a harmonic pencil, namely $(A, B; C, D) = -1$. This suggests we may perform a projection from this pencil to derive the result.

Since the result is about the midpoint of a segment, we shall consider the point at infinity of the line EF . If we can show that it is the harmonic conjugate of B w.r.t. E, F , then the result follows.

Solution. By [proposition 2.4](#), we get $(A, B; C, D) = -1$. Taking A as the perspective centre, we have the transformation

$$\odot (A, B, C, D) \xrightarrow{A} EF(\infty_{EF}, B, E, F).$$

So $(\infty_{EF}, B; E, F) = -1$, and [proposition 2.2](#) shows that $BE = BF$. $\square$



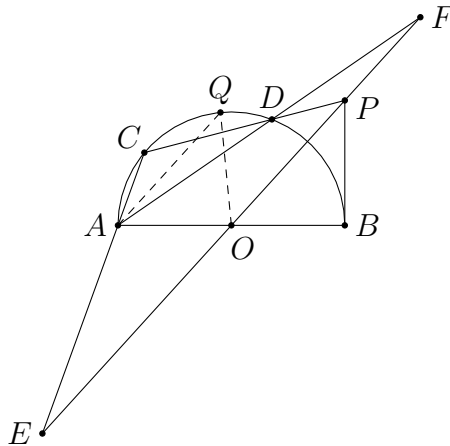
Remarks. Of course one can use similar triangles to solve this problem easily as well. But we should note the simplicity of applying harmonic division. Once we have observed the projection, the problem becomes trivial.

Problem 2.2. (CSMO 2007 Problem 2) Let C, D be two arbitrary points on a semicircle with centre O and diameter AB . The tangent to circle O at the point B intersects the line CD at P . Suppose the line PO meets the lines CA, AD at the points E, F respectively. Prove that $OE = OF$.

Analysis. Similar to [problem 2.1](#), we are trying to prove $(E, F; O, \infty_{EF}) = -1$. By the construction of the points E and F , if we wish to project these points to another pencil, we may use A as the centre. By projecting $EF(E, F, O, \infty_{EF})$ onto the circle, the image of ∞_{EF} is constructed by drawing a parallel line to EF through A . This suggests we should define the point Q in the solution.

It remains to prove $(C, D; B, Q) = -1$. As they are points on the circle, we probably think of [proposition 2.4](#). It means we have to prove that PQ is a tangent to the circle.

Solution.



Let the line through A parallel to EF intersect $\odot O$ again at a point Q . Since

$$\angle POQ = \angle AQO = \angle OAQ = \angle BOP$$

and PB is tangent to $\odot O$, PQ is another tangent to $\odot O$. By [proposition 2.4](#), we know that $(C, D; B, Q) = -1$. Then we can form the perspectivity

$$\odot(C, D, B, Q) \xrightarrow{A} EF(E, F, O, \infty_{EF}).$$

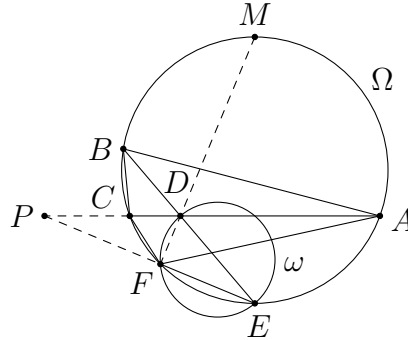
This shows $(E, F; O, \infty_{EF}) = -1$, and hence $OE = OF$ by [proposition 2.2](#). $\square$

Remarks. Note that how harmonic division can be used together with some simple tools in Euclidean geometry.

Problem 2.3. (All-Russian MO 2009 Grade 9 Problem 2) Let BD ($D \in AC$) be the angle bisector in $\triangle ABC$. The line BD intersects the circumcircle Ω of $\triangle ABC$ at B and E . Circle ω with diameter DE intersects Ω again at F . Prove that BF is a symmedian of $\triangle ABC$.

Analysis. By [proposition 2.4](#), this is equivalent to showing that $(A, C; B, F) = -1$. Since any perspective transformation from a pencil on a circle has centre on that circle, a simple way is to project the pencil from E . Also, the right angle on ω suggests the use of [proposition 2.3](#) as well.

Solution.



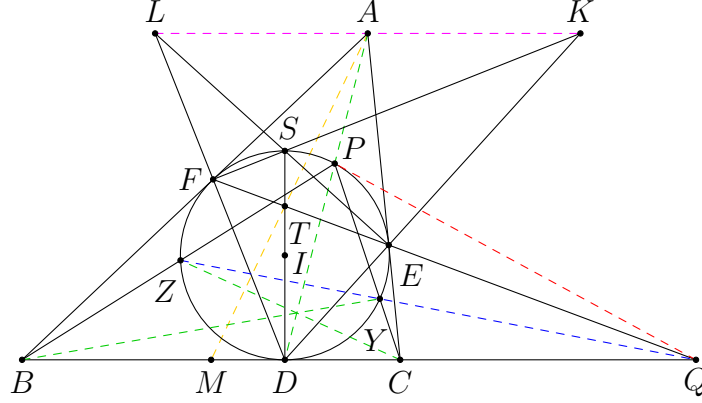
Let $P = EF \cap AC$ and $M = FD \cap \Omega$. Since $\angle EFD = 90^\circ$ and E is the midpoint of $\widehat{AC}$ not containing B , we know that ME is a diameter of Ω and M is the midpoint of $\widehat{ABC}$. In particular, we have $\angle DFC = \angle AFD$. Together with $\angle DFP = 90^\circ$, we have $(A, C; D, P) = -1$ by [proposition 2.3](#). Next,

$$AC(A, C, D, P) \xrightarrow{E} \Omega(A, C, B, F),$$

which implies $(A, C; B, F) = -1$. Then [proposition 2.4](#) shows that BF is a symmedian of $\triangle ABC$. $\square$

In the next problem, we revise some properties involving the incircle. We shall use the new concepts to prove these well-known results.

Problem 2.4. In $\triangle ABC$, let D, E, F be the touching points of the incircle $\odot I$ to the sides BC, CA, AB respectively. Let $P = AD \cap \odot I$, $Q = EF \cap BC$, $Z = BP \cap \odot I$, $Y = CP \cap \odot I$, $S = ID \cap \odot I$, $K = DE \cap FS$, $L = DF \cap ES$, $T = ID \cap EF$ and $M = AT \cap BC$.



Prove the following results.

- (a) QP is a tangent to $\odot I$
- (b) Q, Y, Z are collinear
- (c) BY, CZ, AD are concurrent
- (d) A, K, L are collinear
- (e) $KL \parallel BC$
- (f) (NMO 2010 Problem 6) M is the midpoint of BC

Analysis. As there are many points on the incircle, using Pascal's theorem might help to prove some of the statements. Also, one should pay more attention to the configurations of harmonic division, in particular those involving tangents and mid-points.

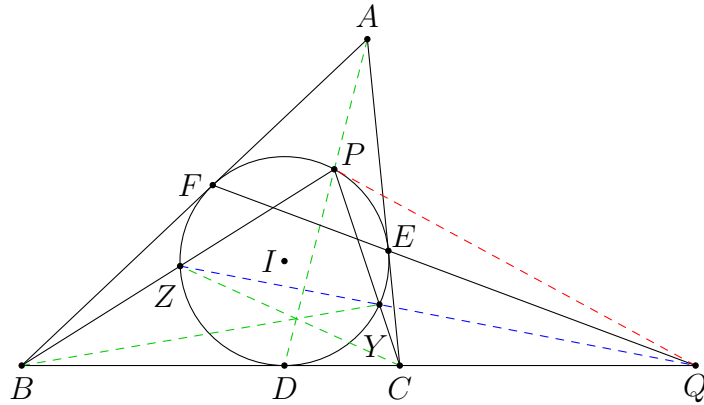
Solution.

- (a) It suffices to prove $(F, E; D, P) = -1$ by [proposition 2.4](#). But this is trivial since the tangents at E and F meet on the line DP at A . $\square$
- (b) It is well-known that AD, BE, CF are concurrent (where the concurrent point is called the [Gergonne point](#)(熱爾崗點) of the triangle). Indeed, one can prove this by a simple application of Ceva's theorem. By [proposition 2.1](#), we get $(B, C; D, Q) = -1$. From the perspectivity

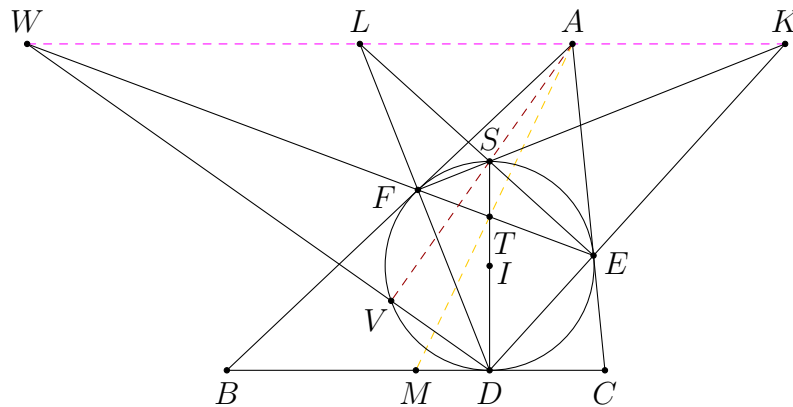
$$BC(B, C, D, Q) \xrightarrow{P} \odot I(Z, Y, D, P),$$

we obtain $(Z, Y; D, P) = -1$. By [proposition 2.4](#) again, this shows the intersection point of the tangents at D and P , which is Q , is collinear with Z and Y . This

9



- 9



- Let $W = EF \cap KL$ and $V = WD \cap \odot I$. It suffices to show $(B, C; M, \infty_{BC}) = -1$. From the definition of the point M , we can only project it to the point T . So we get

$$BC(B, C, M, \infty_{BC}) \xrightarrow{A} EF(F, E, T, W).$$

(This is the reason for defining the point W .) Similarly, to handle the point T , we may project it from D to S or from S to D . Indeed, we choose

$$EF(F, E, T, W) \xrightarrow{D} \odot I(F, E, S, V).$$

It now remains to show $(F, E; S, V) = -1$. As $A = EE \cap FF$, this is equivalent to proving A, S, V are collinear by [proposition 2.4](#). We apply Pascal's theorem again to $SVDEEF$. Then $SV \cap EE, W, K$ are collinear. Since $A = EE \cap WK$, this yields $A = SV \cap EE$, and hence A, S, V are collinear. The result follows readily. $\square$

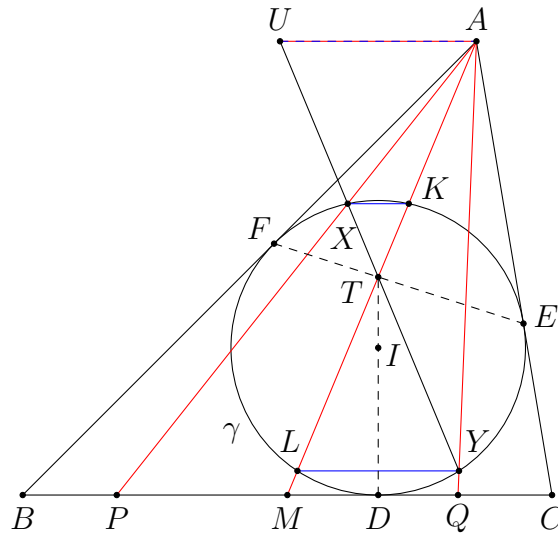
Remarks. These properties about the incircle are quite useful. So it is worthwhile to memorize these as well as the proof. Of course it is always recommended to learn how to use the technique but not simply to follow all details in exactly the same way.

The above tools can also be applied to solve more advanced problems. Sometimes we need to combine knowledge in Euclidean geometry and projective geometry to get useful results.

Problem 2.5. (IMO Shortlist 2005 G6) Let ABC be a triangle, and M the midpoint of its side BC . Let γ be the incircle of triangle ABC . The median AM of triangle ABC intersects the incircle γ at two points K and L . Let the lines passing through K and L , parallel to BC , intersect the incircle γ again in two points X and Y . Let the lines AX and AY intersect BC again at the points P and Q . Prove that $BP = CQ$.

Analysis. The only thing we need to think about when applying projective geometry is how to establish a projective transformation. One usual way is to reverse the order of construction. For example, in this problem, the points are constructed in the order K, L , then X, Y and then P, Q . So our projectivity should follow the reverse order. The remaining steps, including the construction of additional points, are quite natural. To prove that $BP = CQ$, it is the same as proving M is the midpoint of PQ . This recalls a standard configuration of harmonic division.

Solution.



We shall prove that $(P, Q; M, \infty_{BC}) = -1$. Let $T = XY \cap AM$ and $U = XY \cap A\infty_{BC}$ (which means $AU \parallel BC$). Then

$$BC(P, Q, M, \infty_{BC}) \xrightarrow{A} XY(X, Y, T, U) \xrightarrow{\infty_{BC}} AM(K, L, T, A).$$

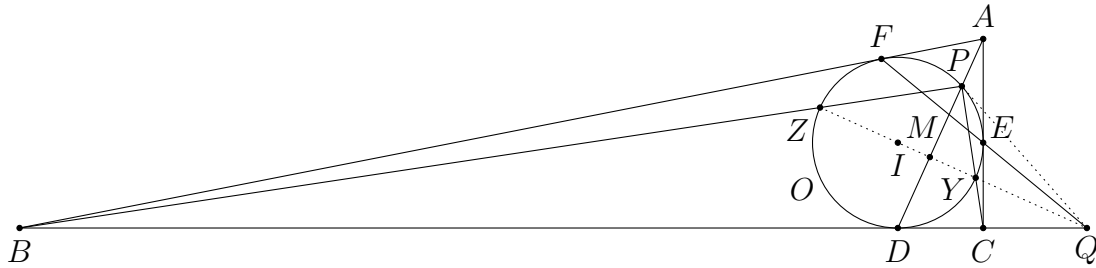
Obviously, the intention of these two steps is to project P, Q back to K, L .

Now, suppose the incircle touches the sides BC, CA, AB at D, E, F respectively. As $LYKX$ is an isosceles trapezoid, its diagonals meet at a point on the axis of symmetry. Thus, all of T, I, D lie on a line perpendicular to BC . By [problem 2.4\(f\)](#), we know that E, T, F are collinear. From this, we obtain $(K, L; T, A) = -1$ by [proposition 2.4](#) as the tangents at E, F intersect at A . It follows from the above projection that $(P, Q; M, \infty_{BC}) = -1$. By [proposition 2.2](#), M is the midpoint of PQ . This implies $BP = CQ$. $\square$

Problem 2.6. (CMO 2006 Problem 4) In a right-angled triangle ABC , $\angle ACB = 90^\circ$. Its incircle O meets BC, CA, AB at D, E, F respectively. AD cuts O at P . If $\angle BPC = 90^\circ$, prove $AE + AP = PD$.

Analysis. From the desired result $AE + AP = PD$ and the fact that ABC is a right-angled triangle, one may guess that the triangle is almost unique, up to a proportional constant, since we have two homogeneous equations in three variables a, b, c , the side lengths of the triangle. Of course this is not always the case, but it seems to be possible to derive the result by computing the lengths. It turns out that this is not an easy task if only the usual tools are used, like Pythagoras' theorem, trigonometric functions, etc. In the following solution, we use some facts in [problem 2.4](#) and harmonic division to handle the conditions.

Solution.



Let $Z = BP \cap O$, $Y = CP \cap O$ and $Q = EF \cap BC$. By [problem 2.4](#), we have the following results.

- QP is tangent to O
- Z, Y, Q are collinear
- $(B, C; D, Q) = -1$

The condition $\angle BPC = 90^\circ$ has two meaningful implications to us. Firstly, it means ZY is a diameter of O , and thus it passes through the incentre I . Secondly, using $(B, C; D, Q) = -1$ and [proposition 2.3](#), we obtain $\angle DPC = \angle CPQ$. These facts are sufficient for us to carry out the computation.

Consider the isosceles $\triangle QPD$, where $QP = QD$ are tangents. Let $QP = QD = x$ and $DP = y$. As $\angle DPC = \angle CPQ$, we apply the angle bisector theorem to obtain $DC = \frac{xy}{x+y}$. Since $IDCE$ is a square, we have $ID = \frac{xy}{x+y}$. Denote by M the midpoint of DP (which lies on IQ). Then $\triangle IDQ \sim \triangle DMQ$ and $\frac{ID}{DQ} = \frac{DM}{MQ}$, which

implies $\frac{y}{x+y} = \frac{\frac{y}{2}}{\sqrt{x^2 - \frac{y^2}{4}}}$. Solving the equation, we obtain the only positive solution

$x = \frac{1+\sqrt{7}}{3}y$. From this relation, we are able to calculate all required ratios.

Indeed, as $(B, C; D, Q) = -1$, we have $\frac{BD}{BQ} = \frac{CD}{CQ} = \frac{y}{x}$. So $BD = \frac{xy}{x-y}$ and

$$BC = BD + DC = \frac{xy}{x-y} + \frac{xy}{x+y} = \frac{2x^2y}{x^2 - y^2}.$$

Recall that $sr = [ABC] = \frac{AC \cdot BC}{2}$, where $s, r, [ABC]$ are the semiperimeter, the inradius and the area of $\triangle ABC$ respectively. This gives

$$(BC + AE) \cdot ID = \frac{(AE + CD) \cdot BC}{2}.$$

Now,

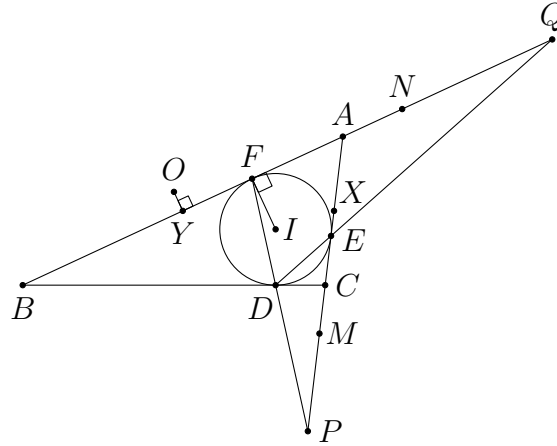
$$\begin{aligned} (BC + AE) \cdot ID &= \frac{(AE + CD) \cdot BC}{2} \\ \Rightarrow 2 \left(\frac{2x^2y}{x^2 - y^2} + AE \right) \cdot \frac{xy}{x+y} &= \left(AE + \frac{xy}{x+y} \right) \cdot \frac{2x^2y}{x^2 - y^2} \\ \Rightarrow \frac{2x^2y}{x^2 - y^2} \cdot \frac{xy}{x+y} &= AE \cdot \left(\frac{2x^2y}{x^2 - y^2} - \frac{2xy}{x+y} \right) \\ \Rightarrow AE &= \frac{x^2}{x+y}. \end{aligned}$$

As $x = \frac{1+\sqrt{7}}{3}y$, we obtain $AE = \frac{2}{3}y$ by simple calculation. Next, using the power of a point theorem, we have $AP \cdot AD = AE^2$, which gives $AP \cdot (AP + y) = \frac{4}{9}y^2$. The only positive solution is $AP = \frac{1}{3}y$. Then $AE + AP = y = PD$ as desired. $\square$

Problem 2.7. (CMO 2007 Problem 4) Let O and I be the circumcentre and the incentre of $\triangle ABC$ respectively. The incircle of $\triangle ABC$ touches the sides BC, CA, AB at the points D, E, F respectively. The lines FD and CA intersect at P , the lines DE and AB intersect at Q . Let M, N be the midpoints of the segments PE, QF respectively. Prove that $OI \perp MN$.

Analysis. We shall use the generalization of Pythagoras' theorem: $WX \perp YZ$ iff $WY^2 - XY^2 = WZ^2 - XZ^2$. The following solution illustrates how the concept of harmonic division simplifies the computation.

Solution.



Let X, Y be the midpoints of AC, AB respectively. Denote by R and r the circumradius and the inradius of $\triangle ABC$ respectively. Note that $OY \perp AB$ and $IF \perp AB$. By Pythagoras' theorem, we have

$$\begin{aligned} ON^2 - IN^2 &= (OY^2 + YN^2) - (IF^2 + FN^2) \\ &= (R^2 - YA^2 + YN^2) - (r^2 + FN^2) \\ &= R^2 - r^2 + (YN - YA)(YN + YA) - FN^2 \\ &= R^2 - r^2 + AN(YN + BY) - FN^2 \\ &= R^2 - r^2 + AN \cdot BN - FN^2. \end{aligned}$$

Since AD, BE, CF are concurrent at the Gergonne point of $\triangle ABC$, we may use [proposition 2.1](#) to get $(B, A; F, Q) = -1$. As N is the midpoint of FQ , we apply [proposition 2.5](#) to get $AN \cdot BN = FN^2$. So $ON^2 - IN^2 = R^2 - r^2$. By symmetry,

$$OM^2 - IM^2 = R^2 - r^2 = ON^2 - IN^2,$$

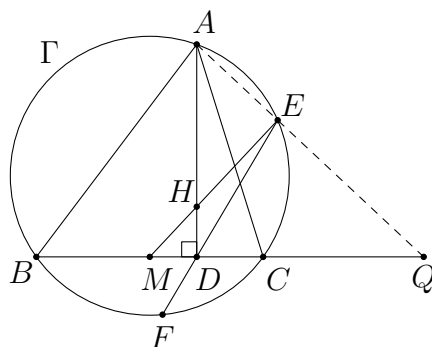
which implies $OI \perp MN$. □

Remarks. One may note that the solution covers all cases as directed lengths can be used.

Problem 2.8. (APMO 2012 Problem 4) Let ABC be an acute triangle. Denote by D the foot of the perpendicular line drawn from the point A to the side BC , by M the midpoint of BC , and by H the orthocentre of ABC . Let E be the point of intersection of the circumcircle Γ of the triangle ABC and the half line MH , and F be the point of intersection (other than E) of the line ED and the circle Γ . Prove that $\frac{BF}{CF} = \frac{AB}{AC}$ must hold.

Analysis. In our new language, the desired result is the same as $(A, F; B, C) = -1$ from the configuration. This suggests using projective geometry. Again, since F is defined as the intersection point of DE and Γ , we can only project F to D using E as the centre (or the opposite). This suggests adding a point $Q = AE \cap BC$. After the projection, it remains to show $(Q, D; B, C) = -1$. This motivates the use of [proposition 2.5](#) since there is a midpoint M of the points B, C in this harmonic division.

Solution.



As A, F are separated by B, C , we know that $\frac{BF}{CF} = \frac{AB}{AC}$ iff $(A, F; B, C) = -1$. Let $Q = AE \cap BC$. Then

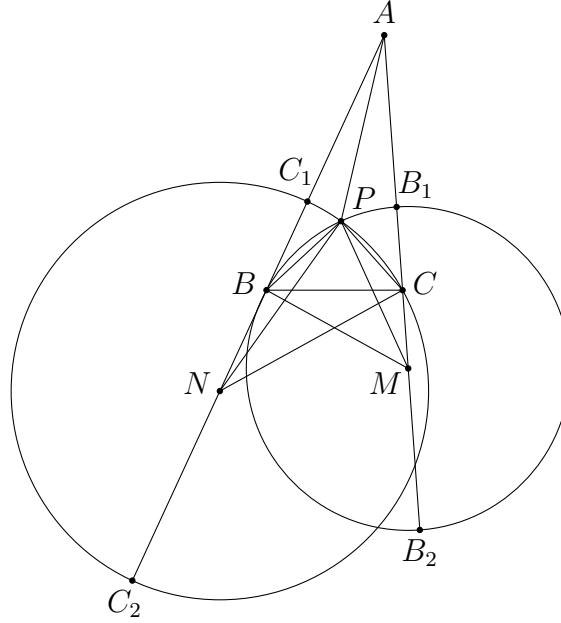
$$\Gamma(A, F, B, C) \xrightarrow{E} BC(Q, D, B, C).$$

Since M is the midpoint of BC , we see from [proposition 2.5](#) that $(Q, D; B, C) = -1$ iff $QC \cdot QB = QD \cdot QM$. Now, $QC \cdot QB = QE \cdot QA$ from the concyclic points A, B, C, E . So it suffices to show $QD \cdot QM = QE \cdot QA$, i.e. A, M, D, E are concyclic. But this is true since $\angle ADM = 90^\circ = \angle AEM$. Note that the last equality is a well-known property of the line MH . $\square$

Problem 2.9. (APMO 2011 Problem 3) Let ABC be an acute triangle with $\angle BAC = 30^\circ$. The internal and external angle bisectors of $\angle ABC$ meet the line AC at B_1 and B_2 , respectively, and the internal and external angle bisectors of $\angle ACB$ meet the line AB at C_1 and C_2 , respectively. Suppose that the circles with diameters B_1B_2 and C_1C_2 meet inside the triangle ABC at point P . Prove that $\angle BPC = 90^\circ$.

Analysis. The configuration consists of some groups of harmonic pencils. So we can use the property of harmonic division to derive some useful results. Though it could be done in the traditional way, this serves as an alternative approach. Sometimes it can save a lot of work.

Solution.



Since $\angle B_1BA = \angle CBB_1$ and $\angle B_2BB_1 = 90^\circ$ (internal and external angle bisectors are perpendicular to each other), we have $(A, C; B_1, B_2) = -1$ by [proposition 2.3](#). Similarly, $(A, B; C_1, C_2) = -1$.

Let M, N be the midpoints of B_1B_2, C_1C_2 respectively. These midpoints are useful because they are the centres of the two circles given in the question.

Now, by [proposition 2.5](#), we have $NB \cdot NA = NC_1^2$. As mentioned, such an equation is closely related to the power of points. Indeed, from

$$NB \cdot NA = NC_1^2 = NP^2,$$

we find that NP is tangent to (ABP) . So $\angle BPN = \angle BAP$. Similarly, note that C also lies on the circle with diameter C_1C_2 because $\angle C_1CC_2 = 90^\circ$. Thus, from $NB \cdot NA = NC_1^2 = NC^2$, we deduce $\angle BCN = \angle BAC$. Due to symmetry, we obtain $\angle CPM = \angle CAP$ and $\angle CBM = \angle CAB$.

Let $\angle BPN = \angle BAP = \alpha$, $\angle CPM = \angle CAP = \beta$ and $\angle NPM = \theta$. Then we have $\angle BCN = \angle CBM = \angle BAC = \alpha + \beta$. Consider the isosceles $\triangle NCP$. The relation $\angle NCP = \angle NPC$ is the same as $\angle BCP + \alpha + \beta = \theta + \beta$, which gives $\angle BCP = \theta - \alpha$. Similarly, $\angle CBP = \theta - \beta$.

Consider $\triangle BCP$. We have $\angle BCP + \angle CBP + \angle BPC = 180^\circ$. This gives

$$(\theta - \alpha) + (\theta - \beta) + (\theta + \alpha + \beta) = 180^\circ,$$

and so $\theta = 60^\circ$. Finally, we obtain $\angle BPC = \theta + \alpha + \beta = 60^\circ + \angle BAC = 90^\circ$. $\square$

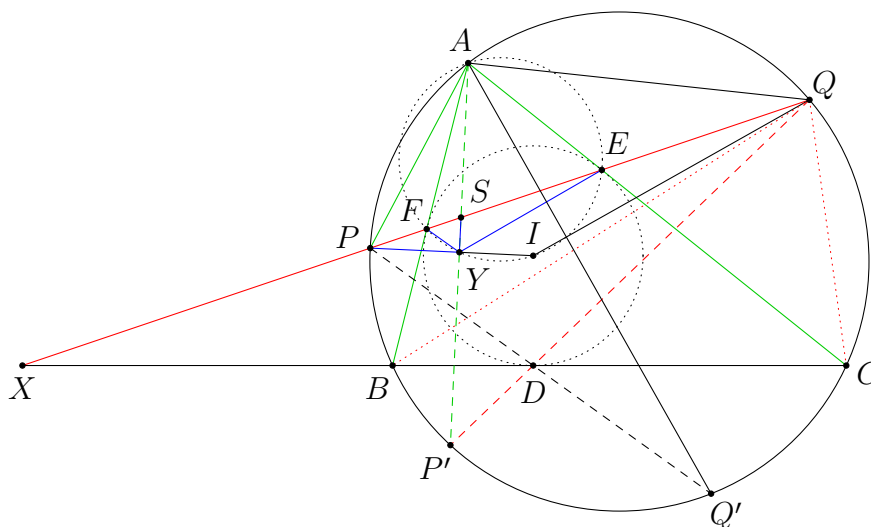
Remarks. From the solution, one notes that the relation $\angle BPC = 60^\circ + \angle BAC$ is always true regardless of the size of $\angle BAC$.

Problem 2.10. (IMO Shortlist 2019 G6) Let I be the incentre of acute-angled triangle ABC . Let the incircle meet BC, CA , and AB at D, E , and F , respectively. Let line EF intersect the circumcircle of the triangle at P and Q , such that F lies between E and P . Prove that $\angle DPA + \angle AQD = \angle QIP$.

Analysis. The first natural idea is to see how to add up $\angle DPA$ and $\angle AQD$ geometrically. This can usually be done if the angles are angles at circumference. Therefore, we may extend PD and QD so that they meet (ABC) again at Q' and P' respectively. After some simple angle chasing, we see that the desired result is the same as $\angle P'AQ' + \angle PIQ = 180^\circ$. Thus, one possible approach is to prove a group of concyclic points. By drawing an accurate figure (which is essential in solving many hard geometry problems), we may notice that $AP' \perp IP$ and $AQ' \perp IQ$. The result follows readily if these hold. Therefore, we are going to prove these perpendicular conditions.

On the other hand, we may look for more properties of the point P' . By using projection (as a possible tool that we may try), we can only project P' to D with centre Q . Thus, we can project some more points on (ABC) to the line BC using Q as the centre and see what happens. After some consideration, one can show that $(P, P'; B, C) = -1$. With this harmonic division, we can probably link up with the final goal (proving a right angle) using [proposition 2.3](#).

Solution.



Let QD and PD meet (ABC) again at P' and Q' respectively. Let PQ meet BC at X . Since AD, BE, CF are concurrent at the Gergonne point of $\triangle ABC$, we have $(X, D; B, C) = -1$ by [proposition 2.1](#). Since

$$BC(X, D, B, C) \xrightarrow{Q} \odot (P, P', B, C),$$

we have $(P, P'; B, C) = -1$.

Next, let Y be the point on PI such that $AY \perp PI$. In view of our discussion, we are going to prove A, Y, P' are collinear. Let AY meet EF at S . Note that A, F, Y, I, E are concyclic since $\angle AFI = \angle AYI = \angle AEI = 90^\circ$. Also, as $AE = AF$, we know that AY bisects $\angle EYF$. Together with $\angle SYP = 90^\circ$, we have $(P, S; F, E) = -1$ by [proposition 2.3](#). Therefore, the pencils (P, S, F, E) and (P, P', B, C) are projective. Consider

$$EF(P, S, F, E) \xrightarrow{A} \odot (P, *, B, C).$$

The missing point $*$ must be P' . This implies A, S, P' are collinear, hence $AP' \perp IP$.

Similarly, $AQ' \perp IQ$. We easily see that

$$\angle QIP + \angle P'AQ' = 360^\circ - 90^\circ - 90^\circ = 180^\circ.$$

Thus,

$$\angle DPA + \angle AQD = \angle Q'PA + \angle AQP' = \angle Q'P'A + \angle AQ'P' = 180^\circ - \angle P'AQ' = \angle QIP.$$

□

Let us review some standard configurations of harmonic division.

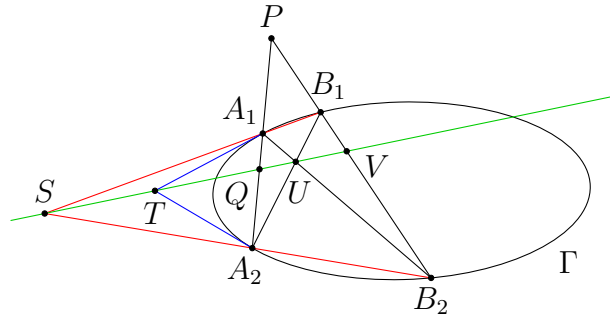
- concurrency and collinearity ([problems 2.4, 2.7, 2.10](#))
- midpoint and point at infinity ([problems 2.1, 2.2, 2.4, 2.5](#))
- right angle and angle bisectors ([problems 2.3, 2.9, 2.10](#))
- tangents and symmedian ([problems 2.1, 2.2, 2.3, 2.4, 2.5, 2.8](#))
- midpoint and power of points ([problems 2.7, 2.8, 2.9](#))

3 Pole and Polar Transformation

We are now going to introduce a transformation in projective geometry, namely, pole and polar transformation. This is a transformation defined with respect to a conic (a circle for our purpose) that interchanges the roles of points and lines. This tool can be used to explain the duality of projective geometry.

Before stating the definition of the transformation, we first consider a conic Γ and any point P . Let ℓ_1, ℓ_2 be some lines through P that cut Γ at $A_1, A_2; B_1, B_2$ respectively. Suppose the tangents at A_1 and A_2 intersect at point T . Applying Pascal's theorem to points $A_1A_1B_1A_2A_2B_2$, we know that $T = A_1A_1 \cap A_2A_2$, $S = A_1B_1 \cap A_2B_2$ and $U = B_1A_2 \cap B_2A_1$ are collinear. Suppose this line cuts A_1A_2 at Q .

Consider $\triangle SA_1A_2$. As collinear points P, B_1, B_2 lie on the three sides of this triangle respectively and SQ, A_1B_2, A_2B_1 are concurrent, we know from [proposition 2.1](#) that $(P, Q; A_1, A_2) = -1$. In other words, Q is a fixed point when ℓ_2 varies, and the line TQ depends at most on the line ℓ_1 .



Next, define $V = \ell_2 \cap SQ$. We notice from the transformation

$$\ell_1(P, Q, A_1, A_2) \xrightarrow{S} \ell_2(P, V, B_1, B_2)$$

that $(P, V; B_1, B_2) = -1$. So TQ also cuts the harmonic conjugate of P w.r.t. B_1, B_2 . Due to symmetry, we shall obtain the same line TQ by interchanging the roles of ℓ_1, ℓ_2 . That means the line only depends on the point P and the conic Γ . Hence, we have the following definition.

Definition 3.1. Let Γ be a conic and let P be any point. The *polar* (極線) (or *polar line*) of the point P w.r.t. Γ is the line which passes through the harmonic conjugate of P w.r.t. the two intersection points of Γ and any line through P .

This definition is well-defined even when P lies on or inside Γ . Dually, we can define the pole of any line ℓ w.r.t. Γ . From the construction, one may notice that the polar line of the pole of ℓ w.r.t. Γ is ℓ itself. Therefore, we may define this term as follows.

Definition 3.2. Let Γ be a conic and let ℓ be any line. The *pole* (極點) of the line ℓ w.r.t. Γ is the unique point P whose polar line w.r.t. Γ is ℓ .

In the remaining context, we shall restrict our attention to circles. We now state the useful results regarding poles and polars.

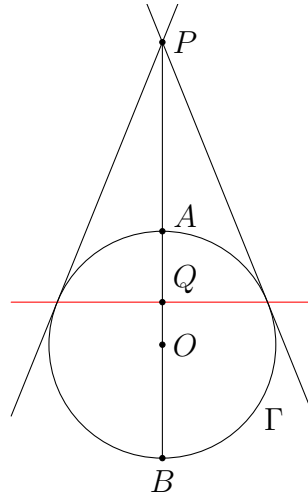
Proposition 3.1. Let Γ be a circle with centre O and radius r . Then the following hold.

- (a) The polar of a point at infinity is the line through O perpendicular to the direction of that ideal point (conversely, the pole of any line through O is a point at infinity).
- (b) The polar of the point O is the line at infinity (conversely, the pole of the line at infinity is O).
- (c) For an ordinary point $P \neq O$, let Q be the point on the ray OP such that

$$OP \cdot OQ = r^2.$$

Then the polar of P is the line through Q perpendicular to OP .

Proof.

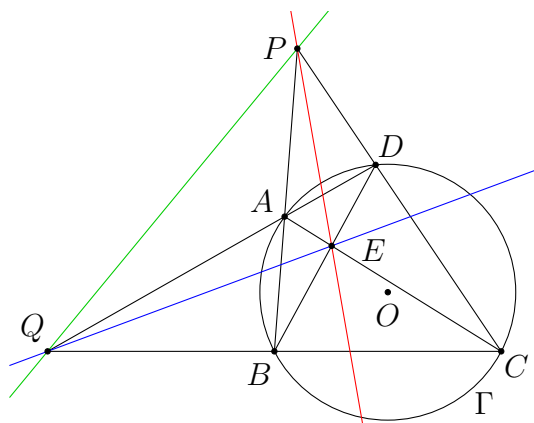


We shall prove only (c) as the others are trivial. Due to symmetry, the polar of P is perpendicular to OP . Let the line OP cut Γ at A, B . Since O is the midpoint of AB , the condition $OP \cdot OQ = r^2 = OA^2$ implies $(P, Q; A, B) = -1$ by [proposition 2.5](#). The result follows from the definition of polar. $\square$

Note that Q is indeed the inversion of P w.r.t. Γ . Thus, it is possible to define the polar of P to be the line perpendicular to OP that passes through the inversion image of P w.r.t. Γ . This is the usual definition of pole and polar transformation for circles in Euclidean geometry, though this does not generalize to an arbitrary conic.

When P lies outside the circle, the polar of P passes through the contact points of tangents from P . This gives one useful way to find the polar of P . Also, when P lies on the circle, its polar is the tangent at P .

Theorem 3.1. (Brokard's theorem(勃羅卡定理)) Let A, B, C, D be points on a circle Γ with centre O . Let $E = AC \cap BD$, $P = AB \cap CD$ and $Q = BC \cap DA$ where all points are ordinary points. Then the polar of each vertex of $\triangle EPQ$ is the line joining the other two vertices. In particular, each of O, E, P, Q is the orthocentre of the triangle formed by the other three points.



The result simply follows from the construction of the polar line and [proposition 3.1](#). One may refer to the first figure given in this section.

The configuration in Brokard's theorem occurs frequently. The collection of all four vertices of a quadrilateral together with the six lines joining pairs of vertices is called a [complete quadrangle](#)(完全四線形), while the triangle formed by the three intersections of opposite sides (the two diagonals are considered as opposite sides) of a complete quadrangle is called its [diagonal triangle](#) (e.g. $\triangle EPQ$).

Note that Brokard's theorem implies the useful results $OE \perp PQ$, $OP \perp EQ$ and $OQ \perp EP$.

Theorem 3.2. (La Hire's theorem(拉伊爾定理)) The polar of P w.r.t. a circle Γ passes through Q iff the polar of Q w.r.t. Γ passes through P .

This is a simple consequence of Brokard's theorem. One may deduce the result by constructing an arbitrary line AD through Q .

By this important theorem, we see if two lines intersect at some point P , then the line joining the poles of the two lines will be the polar of P . Usually, we use this fact to locate the images of points or lines under a pole and polar transformation, as this is much simpler than using the definition or [proposition 3.1](#).

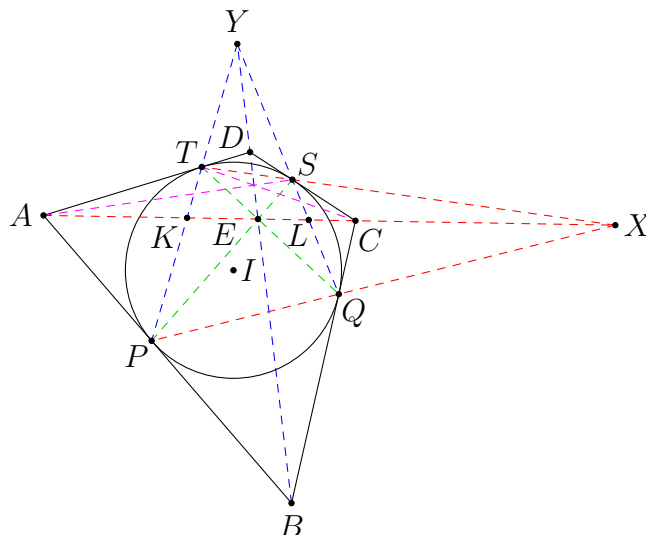
Problem 3.1. Let $ABCD$ be a circumscribed quadrilateral. Its incircle $\odot I$ touches the sides AB, BC, CD, DA at P, Q, S, T respectively. Let $E = AC \cap BD$, $X = PQ \cap ST$, $Y = QS \cap TP$, $K = AC \cap TP$ and $L = AC \cap QS$. Prove the following results.

- (a) AC, PQ, ST are concurrent at X (similarly, BD, QS, TP are concurrent at Y)
- (b) AC, BD, PS, QT are concurrent at E
- (c) $(A, C; E, X) = -1$
- (d) $(K, L; E, X) = -1$
- (e) BD, AS, CT are concurrent

Analysis. Note that (a) and (b) have been proved in [problem 1.6](#). This time we shall use the new tool just introduced to establish these facts. All other facts are equally useful and the configuration of circumscribed quadrilateral is quite common when applying the pole and polar transformation. One should memorize these facts.

Solution.

- (a) Firstly, [proposition 3.1](#) shows that PT is the polar of A w.r.t. $\odot I$. As Y lies on PT , La Hire's theorem implies A lies on the polar of Y . Similarly, C lies on the polar of Y . Next, Brokard's theorem shows that X lies on the polar of Y . Therefore, X lies on AC . Equivalently, AC, PQ, ST are concurrent at X . Similarly, Y lies on BD . $\square$
- (b) Secondly, Brokard's theorem also shows that $PS \cap QT$ lies on the polar of Y , which is the line AC . Then AC, PS, QT are concurrent. Similarly, BD also passes through the concurrent point as desired. $\square$



(c) Recall from the definition of the polar line that $(P, T; K, Y) = -1$. Using

$$PT(P, T, K, Y) \xrightarrow{A} BD(B, D, E, Y),$$

we see that $(B, D; E, Y) = -1$. Due to symmetry, we have $(A, C; E, X) = -1$. $\square$

(d) We have

$$AC(A, C, E, X) \xrightarrow{D} ST(T, S, BD \cap ST, X) \xrightarrow{Y} AC(K, L, E, X).$$

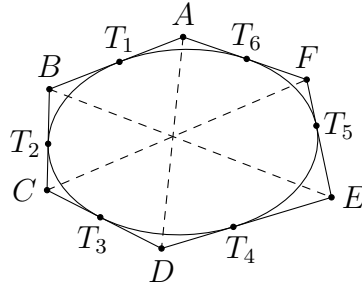
This gives $(K, L; E, X) = -1$. $\square$

(e) Consider $\triangle DAC$. As $(A, C; E, X) = -1$ and X, S, T are collinear, we use [proposition 2.1](#) to deduce DE, AS, CT are concurrent. $\square$

Problem 3.2. ([Brianchon's theorem](#)) Let $ABCDEF$ be a hexagon. Its main diagonals AD, BE, CF are concurrent iff it is circumscribed about a conic.

Analysis. Recall that Brianchon's theorem is the dual of Pascal's theorem. We are going to provide a proof based on Pascal's theorem, with the use of the pole and polar transformation.

Solution.



Let Γ be the conic touching the lines AB, BC, CD, DE, EF at T_1, T_2, T_3, T_4, T_5 respectively and let AT_6 be the second tangent from A to Γ , with T_6 being the contact point. Consider the pole and polar transformation w.r.t. Γ . Then the polar of A, B, C, D, E are $T_6T_1, T_1T_2, T_2T_3, T_3T_4, T_4T_5$ respectively (this is a generalization of [proposition 3.1](#)).

Applying Pascal's theorem to $T_1T_2T_3T_4T_5T_6$, we know that

$$X = T_1T_2 \cap T_4T_5, Y = T_2T_3 \cap T_5T_6, Z = T_3T_4 \cap T_6T_1$$

are collinear, lying on some line ℓ . By La Hire's theorem, X and Z are the poles of BE and AD respectively. Also, ℓ is the polar of $BE \cap AD$.

Now,

- AD, BE, CF are concurrent
- $\Leftrightarrow AD \cap BE, C, F$ are collinear
- $\Leftrightarrow$ the polars of $AD \cap BE, C, F$ are concurrent (by La Hire's theorem)
- $\Leftrightarrow$ the polar of F passes through $\ell \cap T_2T_3$
- $\Leftrightarrow$ the polar of F passes through T_6
(as $Y = T_2T_3 \cap T_5T_6$ lies on ℓ , and the polar of F passes through T_5)
- $\Leftrightarrow AF$ is a tangent to Γ
- $\Leftrightarrow ABCDEF$ is circumscribed about a conic.

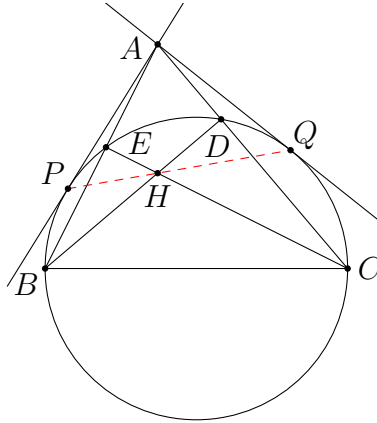
The result is established. $\square$

Remarks. In some sense, one may see that the pole and polar transformation is one of the tools which helps explain the principle of duality. So it is quite an important transformation in the projective plane.

Problem 3.3. (CMO 1996 Problem 1) Let H be the orthocentre of an acute $\triangle ABC$. Construct the tangents AP, AQ from A to the circle with BC as a diameter, where P, Q are the contact points. Prove that P, H, Q are collinear.

Analysis. Using the facts introduced, it is quite obvious that all three points lie on the polar of A w.r.t. the circle. This problem simply shows the usage of pole and polar transformation. Usually, this is used to show that some points are collinear (or some lines are concurrent) by investigating the polar lines of some points (correspondingly, the pole of some lines).

Solution.



By [proposition 3.1](#), PQ is the polar of A w.r.t. the circle with diameter BC . Let D, E be the feet of altitudes from B, C respectively. Then D, E lie on the circle as $\angle BDC = \angle BEC = 90^\circ$. So H lies on the polar of A by Brokard's theorem. This concludes P, H, Q are collinear. $\square$

Remarks. Another trivial application of this transformation is the following problem.

(CMO 1997 Problem 4) The quadrilateral $ABCD$ is inscribed in a circle. The extensions of its sides AB and DC intersect at a point P , while the extensions of AD and BC intersect at a point Q . Two tangents QE and QF from Q to that circle are drawn, where E, F are the contact points. Prove that P, E, F are collinear.

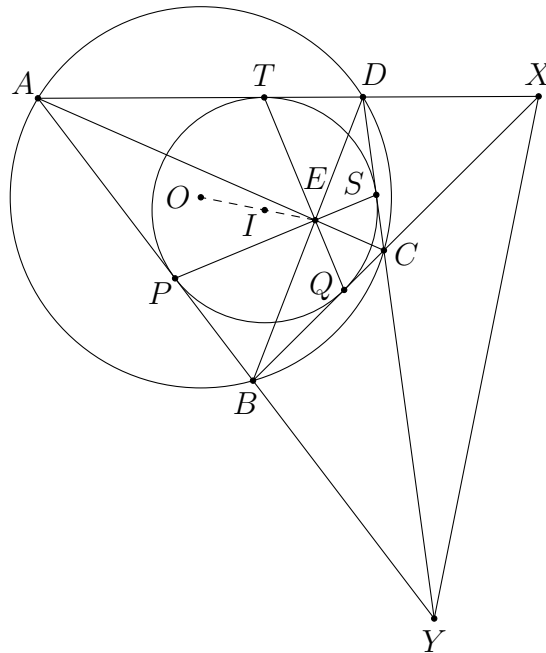
Problem 3.4. Let $ABCD$ be a bicentric quadrilateral. It is inscribed in a circle with centre O and circumscribed about a circle with centre I . Let the diagonals AC and BD meet at E , where O, I, E are pairwise distinct. Prove that O, I, E are collinear.

Analysis. This is the most well-known fact about bicentric quadrilaterals. We demonstrate another way to use pole and polar transformation to prove a collinearity.

If we wish to prove that all of O, I, E lie on the polar line of some point, then such a point must be a point at infinity because O and I are the centres of the only circles in the figure. Thus, this approach does not work.

Indeed, it is also useful to consider the polar lines w.r.t. different circles. For example, as O and I are centres of two circles, one way is to prove that the polar lines of E w.r.t. the two circles are parallel.

Solution.



Let $X = AD \cap BC$ and $Y = AB \cap CD$. By Brokard's theorem, the polar line of E w.r.t. $\odot O$ is the line XY .

Next, let $\odot I$ touch the sides AB, BC, CD, DA at P, Q, S, T respectively. Notice that $E = PS \cap QT$ by [problem 3.1\(b\)](#). As the poles of the lines PS, QT w.r.t. $\odot I$ are points Y, X respectively ([proposition 3.1](#)), the polar line of E w.r.t. $\odot I$ is again XY by La Hire's theorem.

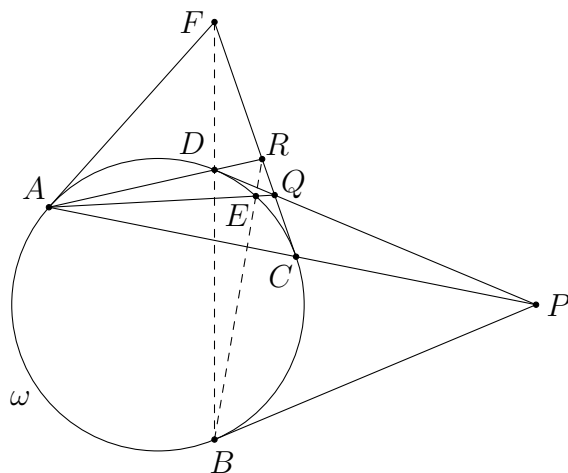
Then by [proposition 3.1](#), both OE and IE are perpendicular to XY . Thus, O, I, E are collinear. $\square$

Problem 3.5. (APMO 2013 Problem 5) Let $ABCD$ be a quadrilateral inscribed in a circle ω , and let P be a point on the extension of AC such that PB and PD are tangent to ω . The tangent at C intersects PD at Q and the line AD at R . Let E be the second point of intersection between AQ and ω . Prove that B, E, R are collinear.

Analysis. As the problem mainly involves collinearity of points and concurrency of lines (and a circle), one may consider Pascal's theorem. To apply this theorem, the line of Pascal should contain three intersection points of secants which do not lie on the circle. So an additional point has to be added.

An ideal way is to add a point which is the intersection of some lines. Since many tangents are involved, we may think of the facts in the pole and polar transformation or harmonic division. For example, one may note that the tangents at A and C meet on the line BD . With this fact, the solution is rather simple.

Solution.



Let F be the intersection of the tangents at A and C . By [proposition 3.1](#), the polar of F w.r.t. ω is AC . Since P lies on this line, the polar of P , which is BD , should pass through F by La Hire's theorem. Alternatively, one may prove this by observing that $(A, C; B, D) = -1$.

Now, applying Pascal's theorem to $AADDBE$, the points $F = AA \cap DB$, $AD \cap BE$ and $Q = DD \cap EA$ are collinear. Since $R = FQ \cap AD$, this shows B, E, R are collinear. $\square$

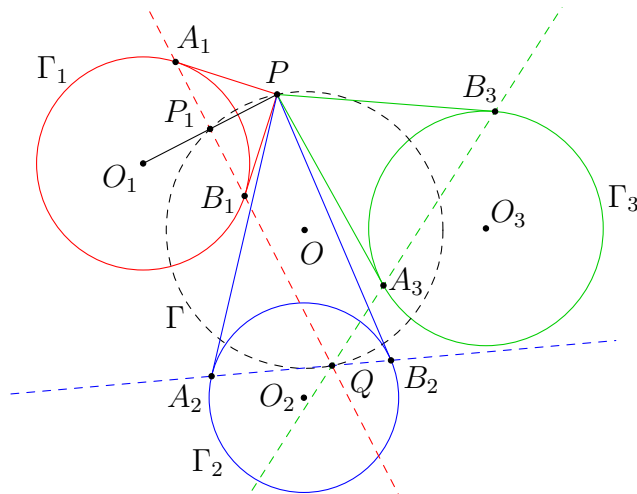
Problem 3.6. (APMO 2009 Problem 3) Let three circles $\Gamma_1, \Gamma_2, \Gamma_3$, which are non-overlapping and mutually external, be given in the plane. For each point P in the plane, outside the three circles, construct six points $A_1, B_1, A_2, B_2, A_3, B_3$ as follows: For each $i = 1, 2, 3$, A_i, B_i are distinct points on the circle Γ_i such that the lines PA_i and PB_i are both tangents to Γ_i . Call the point P *exceptional* if, from the construction, three lines A_1B_1, A_2B_2, A_3B_3 are concurrent. Show that every exceptional point of the plane, if exists, lies on the same circle.

Analysis. Each of the lines A_iB_i is the polar line of P w.r.t. Γ_i . Although we do not have many facts relating the polars of a point w.r.t. different circles, our knowledge provides sufficient information that we need.

Firstly, the concurrent point Q of A_1B_1, A_2B_2, A_3B_3 lies on the polars of P w.r.t. the three circles. So its polars w.r.t. the three circles should all pass through P , and hence are concurrent. In other words, P and Q should lie on the same circle which is to be determined. Due to this symmetry, it is natural to guess PQ is a diameter of the desired circle. In particular, the midpoints of A_1B_1, A_2B_2, A_3B_3 should lie on this circle. The solution is based on these facts.

Indeed, if one is familiar with plane geometry, it is not hard to guess what the desired circle is. As the circle should be symmetric to the three given circles and is unique, a natural choice is the unique circle which is orthogonal to all three circles. Recall that two circles are orthogonal if they intersect at two distinct points and the tangents at each intersection point to the two circles are perpendicular. The centre of that circle is the radical centre of the three circles, and the square of the radius is the power of the radical centre to any circle (see remark). If one knows this fact, then the solution is even more straightforward.

Solution.



Let O_i and r_i be the centre and radius of Γ_i respectively. Suppose A_1B_1, A_2B_2, A_3B_3 intersect at Q . Consider the circle Γ with PQ as a diameter. By [proposition 3.1](#), A_1B_1 is the polar of P w.r.t. Γ_1 . Also, if we let PO_1 meet A_1B_1 at P_1 , then $O_1P_1 \times O_1P = r_1^2$.

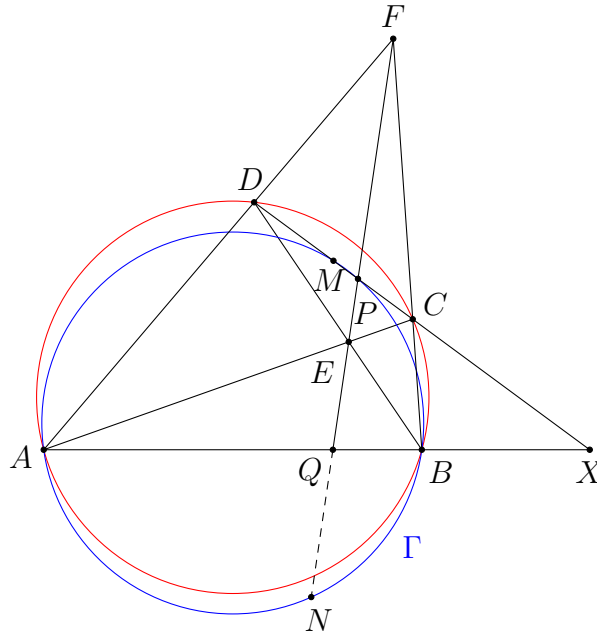
Note that P_1 lies on Γ as $PP_1 \perp P_1Q$. So the power of O_1 w.r.t. Γ is $O_1P_1 \times O_1P = r_1^2$. Recall that the power of O_1 w.r.t. Γ is defined as $OO_1^2 - r^2$, where O and r are the centre and the radius of Γ . By Pythagoras' theorem, this means Γ_1 and Γ are orthogonal to each other. Similarly, Γ is orthogonal to each Γ_i . Hence, Γ is the unique circle which is orthogonal to all three circles, and P always lies on that circle. $\square$

Remarks. Here is a proof of the fact we have used. Suppose Γ is orthogonal to each Γ_i . For Γ and Γ_1 to be orthogonal, we need $r^2 + r_1^2 = OO_1^2$. Similarly, we have $r^2 + r_2^2 = OO_2^2$. Taking the difference of the two equations, we get $OO_1^2 - r_1^2 = OO_2^2 - r_2^2$. This means the powers of O w.r.t. Γ_1 and Γ_2 are equal. So O lies on the radical axis of Γ_1 and Γ_2 . Similarly, O should lie on the radical axis of any two of Γ_i 's, and hence it is the radical centre (if exists) of the three circles. The radius is also uniquely determined by $r = \sqrt{OO_1^2 - r_1^2}$, which is the square root of the power of O w.r.t. Γ_1 .

Problem 3.7. (IMO Shortlist 2004 G8) Given a cyclic quadrilateral $ABCD$, let M be the midpoint of the side CD , and let N be a point on the circumcircle of triangle ABM . Assume that the point N is different from the point M and satisfies $\frac{AN}{BN} = \frac{AM}{BM}$. Prove that the points E, F, N are collinear, where $E = AC \cap BD$ and $F = BC \cap DA$.

Analysis. If one is familiar with harmonic division, one should get $(A, B; N, M) = -1$ at once. So we try to establish a projection between the pencil (A, B, N, M) and another harmonic pencil. In particular, the point N should be obtained through some projections involving the line EF so as to prove that E, F, N are collinear. Indeed, we may start from some familiar harmonic pencil from the definition of polar lines.

Solution.



Let $X = AB \cap CD$, $P = EF \cap CD$, $Q = EF \cap AB$ and let Γ be the circumcircle of $\triangle ABM$. Since EF is the polar of X (Brokard's theorem), we have $(A, B; Q, X) = -1$. Also, $\frac{AN}{BN} = \frac{AM}{BM}$ implies $(A, B; N, M) = -1$ as $N \neq M$ and N, M are separated by A, B on Γ . This shows $AB(A, B, Q, X)$ and $\Gamma(A, B, N, M)$ are projective.

In order to deduce the result P, Q, N are collinear, P should lie on Γ . Once this is done, we simply use P as the centre of perspective transformation between the two pencils to obtain the result. In other words, it suffices to show A, B, P, M are concyclic.

As M is the midpoint of CD , we can use [proposition 2.5](#). Note that $(D, C; P, X) = -1$ by the property of polar line. This implies

$$XP \cdot XM = XC \cdot XD = XB \cdot XA.$$

Thus, A, B, P, M are concyclic. Now, consider the projection

$$AB(A, B, Q, X) \xrightarrow{P} \Gamma(A, B, PQ \cap \Gamma, M).$$

From $(A, B; Q, X) = -1$ and $(A, B; N, M) = -1$, the second intersection of PQ and Γ is exactly N . This shows E, F, N are collinear. $\square$

We summarize the usages of pole and polar transformation below.

- proving concurrency/collinearity ([problems 3.1, 3.3, 3.4, 3.5](#))
- proving perpendicularity ([problem 3.4](#))
- deducing harmonic division pencils ([problems 3.1, 3.7](#))
- deducing dual results ([problem 3.2](#))
- finding power of points ([problem 3.6](#))

4 Desargues' Involution Theorem

In this short section, we introduce a deep theorem. This only has a little use but one could view it as an additional tool to work on some harder problems. Before stating the theorem, we first define the following.

Definition 4.1. Let ℓ be a line on the projective plane. An *involution* (對合) is a function f mapping the points on ℓ to the points on ℓ such that $f(f(A)) = A$ and $(f(A), f(B); f(C), f(D)) = (A, B; C, D)$ for any A, B, C, D on ℓ .

In general, an involution is simply a function f satisfying $f(f(x)) = x$ for any x in its domain. But in our scope, we require f to preserve the cross ratio. One may also define an involution on a conic, but this is not needed for our purpose. For an involution f , if $f(A) = B$, then $f(B) = A$ because of the definition. We say that A and B form a *reciprocal pair* under the involution. The following gives some facts that we need.

Proposition 4.1. Let ℓ be a line. Let f be a function mapping the points on ℓ to the points on ℓ which preserves the cross ratio. Suppose there exist distinct points A and B on ℓ such that $f(A) = B$ and $f(B) = A$. Then f is an involution.

Proof. Consider any other point C on ℓ . Let $f(C) = D$. As f preserves the cross ratio, we have

$$(A, B; C, D) = (f(A), f(B); f(C), f(D)) = (B, A; D, f(D)).$$

Recall that $(B, A; D, f(D)) = (A, B; f(D), D)$. So we obtain

$$(A, B; C, D) = (A, B; f(D), D).$$

By [proposition 1.1\(a\)](#), this implies $C = f(D)$. It follows that $f(f(C)) = f(D) = C$ for any C on ℓ . So f is an involution by definition. $\square$

In other words, to show that f is an involution, it suffices to check that f preserves the cross ratio and find one reciprocal pair under f .

The following fact characterizes all involutions on a line ℓ .

Proposition 4.2. Let ℓ be a line. Let f be a function mapping the points on ℓ to the points on ℓ . Then f is an involution iff it is the restriction of an inversion with some centre P lying on ℓ .

In other words, there exists a point P on ℓ and a nonzero constant k such that $PA \cdot PA' = k$ (in directed lengths) for any ordinary point A on ℓ different from P

where A' is the image of A under the inversion, with the limiting cases $f(P) = \infty_\ell$ and $f(\infty_\ell) = P$.

Proof. Suppose f is an inversion with ratio k . Then we have $f(f(A)) = A$ from the properties of inversion. Consider any points A, B, C, D on ℓ which are different from P and ∞_ℓ . We first show that $(A, B; C, D) = (A', B'; C', D')$ where A', B', C', D' are the images of A, B, C, D under the inversion, i.e.

$$\frac{\frac{AC}{AD}}{\frac{BC}{BD}} = \frac{\frac{A'C'}{A'D'}}{\frac{B'C'}{B'D'}}.$$

Indeed, using the properties of inversion, this holds since

$$\frac{A'C'}{A'D'} \cdot \frac{B'D'}{B'C'} = \frac{\frac{k}{PA \cdot PC} \cdot AC}{\frac{k}{PA \cdot PD} \cdot AD} \cdot \frac{\frac{k}{PB \cdot PD} \cdot BD}{\frac{k}{PB \cdot PC} \cdot BC} = \frac{AC}{AD} \cdot \frac{BD}{BC}.$$

In a similar manner, we can deduce the same result if some of A, B, C, D are P or ∞_ℓ .

Conversely, consider any involution f . Let $P = f(\infty_\ell)$. Then $f(P) = \infty_\ell$. Choose any other point A on ℓ and let $A' = f(A)$. Define $k = PA \cdot PA'$. It suffices to show that $PB \cdot PB' = k$ for any other point B on ℓ where $B' = f(B)$. Indeed, since

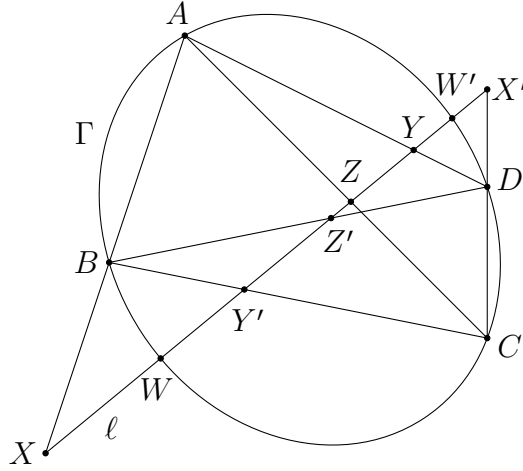
$$(P, \infty_\ell; A, B) = (\infty_\ell, P; A', B'),$$

we have $\frac{PA}{PB} = \frac{PB'}{PA'}$. This yields $PB \cdot PB' = PA \cdot PA' = k$ as desired. $\square$

Note that such an inversion is uniquely determined by two reciprocal pairs, because given points A, B, A', B' on a line, there exists a unique point P on the line such that $PA \cdot PA' = PB \cdot PB'$.

Here comes the core part of our discussion.

Theorem 4.1. (Desargues' involution theorem(笛沙格對合定理)) Let A, B, C, D be distinct points on a conic Γ . Let ℓ be any line not passing through A, B, C, D which intersects Γ at W and W' . Let ℓ meet AB, CD, AD, BC, AC, BD at X, X', Y, Y', Z, Z' respectively. Then there exists an involution on ℓ such that each of (W, W') , (X, X') , (Y, Y') , (Z, Z') is a reciprocal pair.



Proof. By [theorem 1.2](#), there exists a unique function f preserving the cross ratio that maps W, W', X to W', W, X' respectively. By [proposition 4.1](#), since $f(W) = W'$ and $f(W') = W$, f is an involution on ℓ . Thus, $f(X') = X$.

Consider the projection

$$\ell(W, W', X, Y) \xrightarrow{A} \Gamma(W, W', B, D) \xrightarrow{C} \ell(W, W', Y', X').$$

This implies

$$(W, W'; X, Y) = (W, W'; Y', X') = (W', W; X', Y') = (f(W), f(W'), f(X), Y'),$$

and hence we must have $f(Y) = Y'$. This means Y and Y' is a reciprocal pair. Similarly, Z and Z' is a reciprocal pair. $\square$

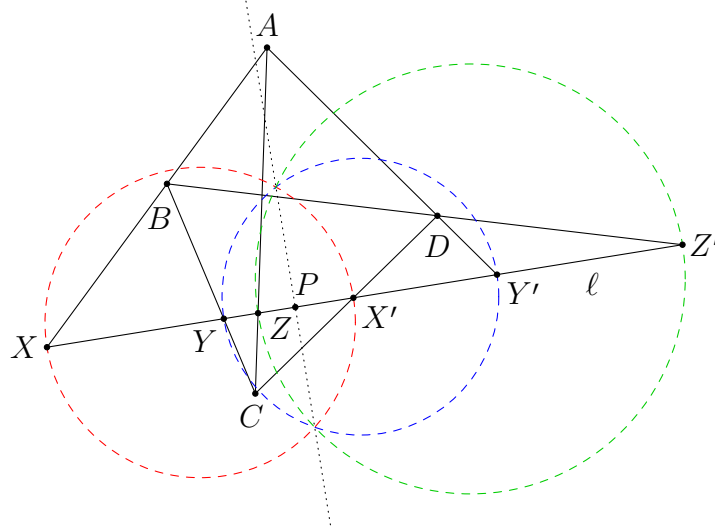
In particular, we mainly deal with the case when A, B, C, D are concyclic, so that Γ is a circle. If there is a line cutting the sidelines and the diagonals of $ABCD$ and cutting the circle, then we may apply Desargues' involution theorem to deduce some reciprocal pairs of an involution. Next, we can use the characterization of involutions to show the existence of a point P such that $PX \cdot PX'$ is a constant, etc.

There are some degenerate cases for Desargues' involution theorem, but we are not going to talk about these.

Problem 4.1. (China TST 2017 Test 2 Problem 3) Let $ABCD$ be a quadrilateral and let ℓ be a line. Let ℓ intersect the lines AB, CD, BC, DA, AC, BD at points X, X', Y, Y', Z, Z' respectively. Given that these six points on ℓ are in the order X, Y, Z, X', Y', Z' , show that the circles with diameters XX', YY', ZZ' are coaxial.

Analysis. The configuration is exactly the same as the one in Desargues' involution theorem, except that a conic is missing. This is not a big deal because we can always find a conic passing through A, B, C, D that intersects ℓ . We shall demonstrate how to use the concept of involution to solve this seemingly hard problem.

Solution.



By Desargues' involution theorem (with an arbitrary conic passing through A, B, C, D and a point on ℓ), we know that there is an involution on ℓ such that $(X, X'), (Y, Y'), (Z, Z')$ are reciprocal pairs. By [proposition 4.2](#), there exists a point P on ℓ such that

$$PX \cdot PX' = PY \cdot PY' = PZ \cdot PZ'.$$

Thus, P has the same power w.r.t. the three circles with diameters XX', YY', ZZ' . Since the centres of these circles lie on the same line ℓ , it is clear that the line through P perpendicular to ℓ is the common radical axis of these circles. This means the circles are coaxial as desired. $\square$

Problem 4.2. (CGMO 2017 Problem 7) Circle ω_1 is the circumcircle of a quadrilateral $ABCD$. The lines AC, BD intersect at point E . The lines AD, BC intersect at point F . A circle ω_2 is tangent to the segments EB, EC at points M, N respectively, and meets the circle ω_1 at points Q, R . The lines BC, AD meet the line MN at points S, T respectively. Prove that Q, R, S, T are concyclic.

Analysis. There is a line which cuts some of the sidelines and diagonals of $ABCD$. This suggests using Desargues' involution theorem to obtain some useful information. In particular, we are able to show the existence of a point P on the line MN satisfying certain power relations, which is crucial to solve this problem. Indeed, we do not need the condition that ω_2 is tangent to BE and CE in the proof below.

Solution. Let MN meet ω_1 at X and Y . By Desargues' involution theorem, there exists an involution on MN such that each of $(X, Y), (M, N), (S, T)$ is a reciprocal pair. By [proposition 4.2](#), there exists a point P on MN such that

$$PX \cdot PY = PM \cdot PN = PS \cdot PT.$$

This section is about some standard usages of Desargues' involution theorem ([problems 4.1, 4.2, 4.3](#)).

Links

Theorems

- [Brianchon's theorem](#)
- [Brokard's theorem](#)
- [Butterfly theorem](#)
- [Cross ratio theorem](#)
- [Desargues' involution theorem](#)
- [Desargues' theorem](#)
- [Fundamental theorem](#)
- [La Hire's theorem](#)
- [Pappus' theorem](#)
- [Pascal's theorem](#)

Propositions

- [1.1](#): Fundamental facts (for points)
- [1.2](#): Fundamental facts (for lines)
- [1.3](#): Intersection of conics and lines
- [2.1](#): Properties of harmonic division 1 (concurrency, collinearity)
- [2.2](#): Properties of harmonic division 2 (midpoint)
- [2.3](#): Properties of harmonic division 3 (angles)
- [2.4](#): Properties of harmonic division 4 (circle)
- [2.5](#): Properties of harmonic division 5 (power)
- [3.1](#): Properties of the pole and polar transformation
- [4.1](#): Criteria for a function to be an involution
- [4.2](#): Characterization of all involutions

Problems

- [1.1](#): Butterfly theorem
- [1.2](#): CSMO 2004 Problem 2
- [1.3](#): Cross ratio
- [1.4](#): Desargues' theorem
- [1.5](#): IMO Shortlist 2006 G2

- 1.6: Pascal's theorem and circumscribed quadrilateral
- 1.7: CHKMO 2011 Problem 2
- 1.8: APMO 2008 Problem 3
- 1.9: IMO Shortlist 2004 G2
- 1.10: CMO 2005 Problem 2
- 1.11: USAMO 2005 Problem 3
- 1.12: IMO Shortlist 2017 G1
- 1.13: IMO 2019 Problem 2
- 2.1: CWMO 2005 Problem 2
- 2.2: CSMO 2007 Problem 2
- 2.3: All-Russian MO 2009 Grade 9 Problem 2
- 2.4: Properties of incircle
- 2.5: IMO Shortlist 2005 G6
- 2.6: CMO 2006 Problem 4
- 2.7: CMO 2007 Problem 4
- 2.8: APMO 2012 Problem 4
- 2.9: APMO 2011 Problem 3
- 2.10: IMO Shortlist 2019 G6
- 3.1: Pole and polar transformation and circumscribed quadrilateral
- 3.2: Brianchon's theorem
- 3.3: CMO 1996 Problem 1
- 3.4: Bicentric quadrilateral
- 3.5: APMO 2013 Problem 5
- 3.6: APMO 2009 Problem 3
- 3.7: IMO Shortlist 2004 G8
- 4.1: China TST 2017 Test 2 Problem 3
- 4.2: CGMO 2017 Problem 7
- 4.3: IMO Shortlist 2012 G8

Terminologies and Notations

- $\ell_1 \cap \ell_2$
- collinear

- complete quadrangle
- concurrent
- conic
- cross ratio (on conics) $(A, B; C, D)$
- cross ratio (on lines) $(A, B; C, D)$
- diagonal triangle
- double ratio $(A, B; C, D)$
- duality
- ellipse
- Gergonne point
- harmonic
- harmonic conjugate
- harmonic division
- hyperbola
- ideal line
- ideal point ∞_ℓ
- involution
- line at infinity
- line conic
- parabola
- pencil of line
- pencil of point (on conics)
- pencil of point (on lines)
- perspective (for conics)
- perspective (for lines)
- perspective (for points)
- perspective (for triangles)
- perspective transformation (for conics)
- perspective transformation (for lines)
- perspective transformation (for points)
- point at infinity ∞_ℓ
- point conic

- polar
- pole
- projective (for conics)
- projective (for points/lines)
- projective transformation
- reciprocal pair
- symmedian
- tangent

References

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(Available from: <https://artofproblemsolving.com/community/q1h1509866p8957048>)
- Theory and Problems of Projective Geometry (by Frank Ayres, JR.)

「前不見古人，後不見來者。念天地之悠悠，獨愴然而涕下。」

《登幽州台歌》